



RESEARCH ARTICLE

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Key Points:

- Unified model simultaneously captures sediment transport threshold and rate data across conditions in water and air within a factor of 2
- Model supports the recent numerical result that the threshold and rate of equilibrium aeolian saltation are insensitive to soil cohesion
- Model challenges the classical prediction that the transport threshold for a longitudinally sloped bed depends on the angle of repose

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Unified Model of Sediment Transport Threshold and Rate Across Weak and Intense Subaqueous Bedload, Windblown Sand, and Windblown Snow

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Abstract Nonsuspended sediment transport (NST) refers to the sediment transport regime in which the flow turbulence is unable to support the weight of transported grains. It occurs in fluvial environments (i.e., driven by a stream of liquid) and in aeolian environments (i.e., wind-blown) and plays a key role in shaping sedimentary landscapes of planetary bodies. NST is a highly fluctuating physical process because of turbulence, surface inhomogeneities, and variations of grain size and shape and packing geometry. Furthermore, the energy of transported grains varies strongly due to variations of their flow exposure duration since their entrainment from the bed. In spite of such variability, we here propose a deterministic model that represents the entire grain motion, including grains that roll and/or slide along the bed, by a periodic saltation motion with rebound laws that describe an average rebound of a grain after colliding with the bed. The model simultaneously captures laboratory and field measurements and discrete element method (DEM)-based numerical simulations of the threshold and rate of equilibrium NST within a factor of about 2, unifying weak and intense transport conditions in oil, water, and air (oil only for threshold). The model parameters have not been adjusted to these measurements but determined from independent data sets. Recent DEM-based numerical simulations (Comola, Gaume, et al., 2019; <https://doi.org/10.1029/2019GL082195>) suggest that equilibrium aeolian NST on Earth is insensitive to the strength of cohesive bonds between bed grains. Consistently, the model captures cohesive windblown sand and windblown snow conditions despite not explicitly accounting for cohesion.

Plain Language Summary Loose sedimentary grains cover much of the wind-blown (i.e., aeolian) and water-worked (i.e., fluvial) sedimentary surfaces of Earth and other planetary bodies. To predict how such surfaces evolve in response to aeolian and fluvial flows, one needs to understand the rate at which sediment is transported for given environmental parameters such as the flow strength. In particular, one needs to know the threshold flow conditions below which most sediment transport ceases. Here, we propose a simple model that unifies most aeolian and fluvial sediment transport conditions, predicting both the sediment transport threshold and rate in agreement with measurements and numerical simulations. Our results will make future predictions of planetary surface evolution more reliable than they currently are.

1. Introduction

When a unidirectional turbulent shearing flow of a Newtonian fluid such as air or water applies a sufficiently strong shear stress τ onto an erodible sediment bed surface, sediment can be transported by the flow (Durán et al., 2011; Garcia, 2008; Kok et al., 2012; Pähtz et al., 2020; Valance et al., 2015). There are two extreme sediment transport regimes: transported grains can enter suspension supported by the flow turbulence and remain out of contact with the bed for very long times, or they can remain in regular contact with the bed (i.e., *nonsuspended*). Fully nonsuspended sediment transport (NST) occurs when the Rouse number $Ro \equiv v_s^- / (\kappa u_*^-)$ exceeds a critical value ($Ro_c \approx 2.8$, Naqshband et al., 2017), where v_s^- is the terminal settling velocity of grains in quiescent flow, $\kappa = 0.4$ the von Kármán constant, and $u_*^- \equiv \sqrt{\tau / \rho_f}$ the fluid shear velocity, with ρ_f the fluid density. The most important examples for NST in nature are the transport of coarse sand and gravel driven by streams of liquids (*fluvial*), such as their transport in rivers of water on Earth (*subaqueous bedload*) and rivers of methane on Saturn's moon Titan (Poggiali et al., 2016), and the

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atmospheric wind-driven (*aeolian*) transport of sand-sized minerals (*windblown sand*) and snow and ice (*windblown snow*).

NST plays a key role in the formation of aeolian and fluvial ripples and dunes on Earth and other planetary bodies (Bourke et al., 2010; Charru et al., 2013). Hence, predicting the morphodynamics of planetary sedimentary surfaces requires a deep physical understanding of NST, especially if predictions are to be made outside the range of conditions that are accessible to measurements, like in extraterrestrial environments (Claudin & Andreotti, 2006; Durán Vinent et al., 2019; Jia et al., 2017; Pähtz et al., 2013; Telfer et al., 2018). A first step toward physically understanding NST in its full complexity is to study its most important statistical properties for idealized situations. Numerous physical studies have therefore focused on predicting the sediment transport rate Q (i.e., the total streamwise particle momentum per unit bed area) for equilibrium (i.e., steady and homogeneous) conditions and a flat bed (i.e., no bedforms) of nearly monodisperse, cohesionless, spherical sedimentary grains of density ρ_p and median diameter d (e.g., Abrahams & Gao, 2006; Ali & Dey, 2017; Ancey, 2020a; Bagnold, 1956, 1966, 1973; Berzi & Fraccarollo, 2013; Berzi et al., 2016; Chauchat, 2018; Doorschot & Lehning, 2002; Durán & Herrmann, 2006; Einstein, 1950; Fraccarollo & Hassan, 2019; Jenkins & Valance, 2014; Lämmel et al., 2012; Matoušek & Zrostlík, 2020; Owen, 1964; Pähtz & Durán, 2020; Pomeroy & Gray, 1990; Sørensen, 1991, 2004; Zanke & Roland, 2020).

Recently, Pähtz and Durán (2020) unified Q across most aeolian and fluvial environmental conditions (non-shallow flows), parametrized by the gravity constant g , bed slope angle α (slopes aligned with flow direction, for downward slopes, $\alpha > 0$), kinematic fluid viscosity ν_f , and ρ_f and τ . These authors first defined the following dimensionless numbers:

$$\text{Density ratio : } s \equiv \rho_p / \rho_f, \quad (1a)$$

$$\text{Galileo number : } Ga \equiv d \sqrt{s \tilde{g}} / \nu_f, \quad (1b)$$

$$\text{Shields number : } \Theta \equiv \tau / (\rho_p \tilde{g} d), \quad (1c)$$

where $\tilde{g} \equiv (1 - 1/s)g \cos \alpha$ is the vertical buoyancy-reduced value of g . The Shields number Θ is a measure for the ratio between tangential flow forces and normal resisting forces acting on bed surface grains (Pähtz et al., 2020), Ga controls the scaling of the dimensionless settling velocity $v_s^- / \sqrt{s \tilde{g} d}$ (Camenen, 2007), while $\sqrt{s}$ separates the settling velocity scale $\sqrt{s \tilde{g} d}$ from the escape velocity scale $\sqrt{\tilde{g} d}$ that grains need to exceed in order to escape the potential traps of the bed surface. Pähtz and Durán (2020) then separated Q into the transport load M (i.e., the total mass of transported grains per unit bed area) and the average streamwise sediment velocity $\bar{v}_x$ via $Q = M \bar{v}_x$ and derived a parametrization for $Q_* \equiv Q / (\rho_p d \sqrt{s \tilde{g} d})$ that incorporates $M_* \equiv M / (\rho_p d)$ and $v_{x*} \equiv \bar{v}_x / \sqrt{s \tilde{g} d}$ (Equation 2b is an improved version of equation (S20) of Pähtz & Durán, 2020, see Appendix A):

$$Q_* = M_* \bar{v}_{x*} (1 + c_M M_*), \quad (2a)$$

$$M_* = (\Theta - \Theta_t) / (\mu_b - \tan \alpha), \quad (2b)$$

where the subscript t refers to threshold conditions, that is, the limit of vanishing dimensionless transport load, $M_* \rightarrow 0$ (i.e., $\Theta \rightarrow \Theta_t$, where Θ_t is the transport threshold).

The parameter μ_b in Equation 2b is the bed surface value of the friction coefficient (i.e., the ratio between particle shear stress and vertical particle pressure), which approximates the ratio between the average streamwise momentum loss and vertical momentum gain of transported grains during their contacts with the bed surface (Pähtz & Durán, 2018b). Furthermore, Equation 2a consists of two additive contributions to Q_* : the term $M_* \bar{v}_{x*}$, which corresponds to the scaling of Q_* in the hypothetical case that collisions between transported grains do not occur, and the term $c_M M_*^2 \bar{v}_{x*}$, which encodes the effect of such collision on Q_* . Using discrete element method (DEM)-based numerical simulations of NST, Pähtz and Durán (2020) found that Equation 2a with $c_M = 1.7$ is universally obeyed across simulated weak and intense equilibrium NST

conditions ($s \in [2.65, 2000]$ and $Ga \in [5, 100]$) that satisfy $s^{1/2} Ga \gtrsim 80$ for $s \lesssim 10$ (typical for fluvial environments) or $s^{1/2} Ga \gtrsim 200$ for $s \gtrsim 10$ (typical for aeolian environments).

The model of Pahzt and Duran (2020) is incomplete because it does not incorporate expressions for Θ_t , μ_b , and $\bar{v}_{x*_{\Theta_t}}$. To overcome this problem, these authors fitted Θ_t to a given experimental or numerical data set, while they used semiempirical closure relations for μ_b and $\bar{v}_{x*_{\Theta_t}}$ from their previous studies: $\mu_b \approx 0.63$ (Pahzt & Duran, 2018b) and $\bar{v}_{x*_{\Theta_t}} \approx 2\kappa^{-1}\sqrt{\Theta_t}$ (limited to $s^{1/4} Ga \gtrsim 40$, Pahzt & Duran, 2018a). However, in the context of general modeling, fitting is undesirable, while semiempirical relationships are problematic when applied to conditions outside the range of the data to which they have been adjusted.

Here, we complete the model of Pahzt and Duran (2020). Instead of relying on fitting of Θ_t and semiempirical closure relations for μ_b and $\bar{v}_{x*_{\Theta_t}}$, we propose a physical transport threshold model that predicts the three unknown quantities Θ_t , μ_b , and $\bar{v}_{x*_{\Theta_t}}$ for conditions with arbitrary s , Ga , and α , unifying NST in oil, water, and air. This threshold model is then coupled with Equations 2a and 2b to predict Q_* . We show that this coupled model simultaneously captures laboratory and field measurements and DEM-based numerical simulations of Θ_t and Q_* within a factor of about 2, though agreement with data of Q_* requires that a critical value of $s^{1/2} Ga$ is exceeded (consistent with the validity limitation of Equation 2a). The only data of Q_* that are not captured by the coupled model (i.e., those with too small $s^{1/2} Ga$) correspond to NST driven by viscous liquids such as oil (Charru et al., 2004).

From the threshold model predictions, we also derive simple scaling laws for Θ_t valid for different NST regimes. Special attention is paid to the predicted effect of α on Θ_t for these regimes, which we compare with the classical bed slope correction of Θ_t .

A further aspect that is addressed in this study is the effect of soil cohesiveness on Θ_t and Q_* . From DEM-based numerical simulations of equilibrium aeolian NST for Earth's atmospheric conditions, Comola, Gaume, et al. (2019) suggested that the strength of cohesive bonds between bed grains does neither significantly affect Θ_t nor Q_* even though it strongly affects the transient toward the equilibrium. However, this suggestion is very controversial. If it was indeed generally true, it would indicate a conceptual problem in most, if not all, existing aeolian transport threshold models. In fact, existing threshold models, regardless of whether they model transport initiation (e.g., Burr et al., 2015, 2020; Iversen & White, 1982; Lu et al., 2005; Shao & Lu, 2000) or transport cessation (e.g., Andreotti et al., 2021; Berzi et al., 2016; Claudin & Andreotti, 2006; Kok, 2010; Pahzt et al., 2012), usually incorporate expressions that describe the entrainment of bed surface grains by the flow and/or grain-bed impacts, and both entrainment mechanisms are strongly hindered by cohesion. Furthermore, even those few existing models that do not consider bed sediment entrainment explicitly account for cohesive forces (Berzi et al., 2017; Pahzt & Duran, 2018a), which increase the calculated transport threshold.

Here, using numerical data provided by Comola, Gaume, et al. (2019), we show that, in contrast to previous threshold models, the conceptualization behind our threshold model supports the suggested insensitivity of aeolian NST to cohesion and propose a criterion for when cohesion can be expected to become important. Consistently, we validate the cohesionless coupled model with transport threshold and rate data not only for cohesionless aeolian and fluvial conditions but also for cohesive aeolian conditions, including aeolian NST of small mineral grains and of potentially very cohesive snow grains.

The remainder of the paper is organized as follows. Section 2 presents the transport threshold model, Section 3 the results, such as the evaluation of the coupled model with existing experimental and numerical data of transport threshold and rate, Section 4 the discussion of the results, and Section 5 conclusions drawn from it. Furthermore, there are several appendices, containing lengthy justifications of model assumptions and mathematical derivations, and a notations section at the end of the paper.

2. Transport Threshold Model

NST is a highly fluctuating physical process (Ancy, 2020b; Duran et al., 2011) because of turbulence, surface inhomogeneities, and variations of grain size and shape and packing geometry. Furthermore, the energy of transported grains varies strongly due to variations of their flow exposure duration since their

entrainment from the bed. In fact, grains entrained by the flow initially roll and exhibit a comparably very low kinetic energy, while they can saltate in large hops (especially for aeolian NST), associated with a comparably very large kinetic energy, once they have survived a sufficient number of grain-bed interactions. In spite of such variability, we make the following idealizations to model the transport threshold Θ_t , the bed friction coefficient μ_b , and the threshold value of the average streamwise particle velocity $\overline{v_{x*}}_t$:

1. The mean motion of grains driven by a fluctuating turbulent flow along an inhomogeneous bed surface is represented by the motion of spherical, monodisperse grains driven by the mean (i.e., nonfluctuating) turbulent flow along the mean bed surface (i.e., grain-bed interactions are represented by their statistical mean effect). A partial justification of this idealization is presented in Appendix B
2. Given the idealization above, Θ_t is defined as the smallest Shields number for which a nontrivial steady state grain trajectory exists. This definition of Θ_t is conceptually equivalent to the one of Almeida et al. (2006, 2008). It implies that, for $\Theta < \Theta_t$, all transported grains lose more kinetic energy during their rebounds with the bed than they gain during their hops and thus eventually settle
3. The threshold steady state trajectory is modeled as a periodic saltation trajectory, where grain-bed interactions are described as the average grain-bed rebound, even for NST regimes in which a significant or predominant portion of grains roll and/or roll slide along the bed. However, for this trajectory to be consistent with a sustained rolling motion, we require that grains following this trajectory exhibit a sufficient kinetic energy to be able to roll (or hop) out of the most stable pockets of the bed surface assisted by the near-surface flow. A justification of this idealization is presented in Appendix C. It is partially based on a bed friction law associated with general equilibrium NST, which is introduced in Section 2.2
4. The quantities μ_b and $\overline{v_{x*}}_t$ are calculated from the periodic saltation trajectory

In the following subsections, we derive step-by-step the transport threshold model. First, we present basic assumptions that characterize flow, particles, and their interactions (Section 2.1). Second, we introduce the bed friction law that Equation 2b is based on and show that it leads to an expression linking the average difference between fluid and grain velocity to the bed friction coefficient μ_b (Section 2.2). This friction law, when combined with insights from previous DEM-based numerical simulations of NST, supports representing the entire grain motion in equilibrium NST by grains saltating in identical periodic trajectories with rebound boundary conditions (Appendix C). Third, we present the mathematical description of this periodic saltation motion (Section 2.3). Fourth, we present the manner in which Θ_t , μ_b , $\overline{v_{x*}}_t$, and the equilibrium dimensionless sediment transport rate Q_* are obtained from the family of identical periodic trajectory solutions (Section 2.4).

2.1. Basic Assumptions

2.1.1. Flow Velocity Profile

We consider a mean inner turbulent boundary layer flow above the bed (justified in Appendix B) and assume that this flow is undisturbed by the presence of transported grains, since the nondimensionalized mass of transported sediment per unit bed area M_* becomes arbitrarily small (i.e., $M_* \rightarrow 0$) when Θ approaches Θ_t from above (i.e., $\Theta \rightarrow \Theta_t$) because of Equation 2b. The inner turbulent boundary layer is defined by a nearly height-invariant total fluid shear stress in the absence of transported grains (i.e., $d\tau/dz \simeq 0$, George, 2013). This definition implies that the boundary layer thickness or flow depth is much larger than the transport layer thickness (i.e., NST driven by water flows with relatively small flow depth, like in mountain streams, is excluded). The streamwise flow velocity profile $u_x(z)$ within the inner turbulent boundary layer (the *law of the wall*, Smits et al., 2011) is controlled by the fluid shear velocity u_* and the shear Reynolds number $Re_d \equiv u_* d / \nu_f = Ga \sqrt{\Theta}$. The law of the wall exhibits three regions: a log layer, $u_x = \kappa^{-1} u_* \ln[30(z/d + Z_\Delta)]$, for large wall units $Re_z \equiv Re_d(z/d + Z_\Delta)$; a viscous sublayer, $u_x = u_* Re_z$, for small Re_z ; and a buffer layer for intermediate Re_z ; where $Z_\Delta \equiv z_\Delta/d = 0.7$, with $-z_\Delta$ the virtual zero level of u_x below the bed surface elevation $z = 0$ (defined in Figure 1).

The latter two layers vanish when the bed surface becomes too rough (i.e., $Re_d \gtrsim 70$). Although it is sometimes conjectured that NST breaks up the viscous sublayer (Kok et al., 2012; White, 1979), DEM-based numerical simulations of NST suggest that this is actually not the case (Durán et al., 2011, Figure 22). In particular, any potential effect from moving grains should vanish in the limit of threshold conditions

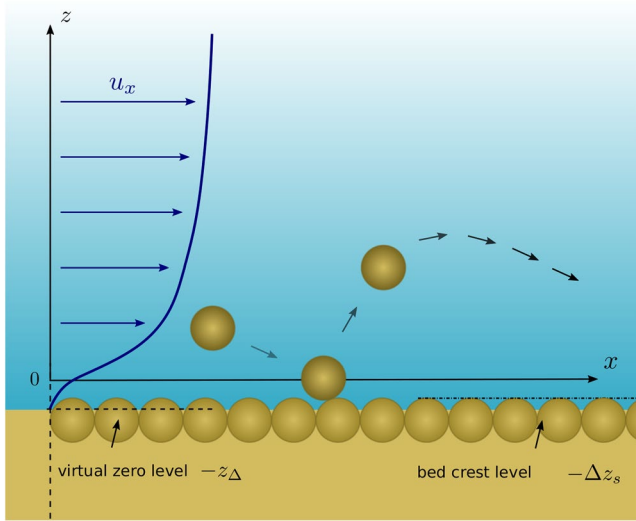


Figure 1. Sketch visualizing the definition of the bed surface. The bed surface is defined as the elevation at which $p_g d \langle v_x \rangle / dz$ is maximal and assigned the vertical coordinate $z = 0$, where p_g is the vertical particle pressure, $\mathbf{v}$ the grain velocity, and $\langle \cdot \rangle$ denotes a local grain mass-weighted ensemble average (Pähtz & Durán, 2018b). The quantity $\frac{1}{2} p_g d \langle v_x \rangle / dz$ describes the production rate of the cross-correlation fluctuation energy density $-\rho \langle v'_z v'_x \rangle$, where ρ is the particle concentration and $\mathbf{v}' \equiv \mathbf{v} - \langle \mathbf{v} \rangle$ the fluctuation grain velocity (Pähtz & Durán, 2018b). Since the main source of production of $-\rho \langle v'_z v'_x \rangle$ are grain-bed rebounds, the bed surface elevation defined in this manner corresponds to the effective elevation of the center of mass of energetic grains when they rebound from the bed surface (Pähtz & Durán, 2018b). In DEM-based numerical simulations of NST, this elevation is about $z_\Delta = 0.7 d$ above the virtual zero level of u_x near threshold conditions (Pähtz & Durán, 2018a). This value is consistent with measurements of the distance $\Delta z_s \approx 0.25 d$ between the crest level of static bed grains and the virtual zero level of u_x in both laminar (Hong et al., 2015) and turbulent (Dey et al., 2012) flows. DEM, discrete element method; NST, nonsuspended sediment transport.

(i.e., nonfluctuating) fluvial conditions ($\tau^{\text{fluc}} = \tau$), such a bed surface is able to resist all driving flows with Shields numbers $\Theta < \Theta_Y$ (Pähtz et al., 2020). The so-called *yield stress* Θ_Y is therefore a statistical quantity encoding the resistance of the bed surface as a whole, though it may be interpreted as the nondimensionalized fluctuating fluid shear stress $\Theta^{\text{fluc}} \equiv \tau^{\text{fluc}} / (\rho_p \tilde{g} d)$ that is required to initiate rolling of grains resting in the most stable pockets of the bed surface (Clark et al., 2017). For a nonsloped bed of nearly monodisperse, cohesionless, frictional spheres, $\Theta_Y^o \equiv \Theta_Y|_{\alpha=0}$ is expected to exhibit a universal value (Pähtz et al., 2020). Based on measurements for laminar fluvial driving flows (Charru et al., 2004; Houssais et al., 2015; Loiseleux et al., 2005; Ouriemi et al., 2007), we use the approximate value $\Theta_Y^o = 0.13$.

2.2. Bed Friction Law

The transport threshold model derivation starts with describing general equilibrium NST by a bed friction law that goes back to Bagnold (1956); Bagnold (1966, 1973), and which also led to the derivation of Equation 2b. In fact, the bed friction coefficient μ_b in Equation 2b is rigorously linked to the streamwise (a_x) and vertical (a_z) components of the acceleration $\mathbf{a}$ of transported grains due to noncontact forces via (Pähtz & Durán, 2018a; Pähtz & Durán, 2020)

$$\mu_b = -\overline{a_x} / \overline{a_z}, \quad (4)$$

because of $M_* \rightarrow 0$. We use the parametrization of the law of the wall by Guo and Julien (2007), which covers the entire ranges of Re_z and Re_d in a single expression:

$$u_x = u_* f_u(Re_d, z/d) = u_* f_u(Re_d, Re_z), \text{ with} \\ f_u(Re_d, Re_z) \equiv 7 \arctan\left(\frac{Re_z}{7}\right) + \frac{7}{3} \arctan^3\left(\frac{Re_z}{7}\right) - 0.52 \arctan^4\left(\frac{Re_z}{7}\right) \\ + \ln\left[1 + \left(\frac{Re_z}{B_\kappa}\right)^{(1/\kappa)}\right] - \frac{1}{\kappa} \ln\left[1 + 0.3 Re_d \left[1 - \exp\left(-\frac{Re_d}{26}\right)\right]\right], \quad (3)$$

where $B_\kappa \equiv \exp(16.873\kappa - \ln 9)$.

2.1.2. Fluid-Particle Interactions

Like recent numerical studies of the physics of aeolian and fluvial sediment transport (e.g., Durán et al., 2012; Schmeckle, 2014), we consider the fluid drag and buoyancy forces as fluid-particle interactions but neglect other interaction forces because (i) they are usually much smaller than the drag force for grains in motion, (ii) there is no consensus about how these forces behave as a function of the distance from the bed surface, and (iii) we are only looking for the predominant effect and are content with an agreement between model and experimental data within a factor of 2. Details are explained in Appendix D.

2.1.3. Sedimentary Grains and Sediment Bed

We consider a random close packed bed made of nearly monodisperse, cohesionless, spherical sedimentary grains. Furthermore, we assume that flows have worked on this bed for a sufficiently long time such that it has reached a state of maximum resistance (Clark et al., 2017; Pähtz et al., 2020). In this state, a quiescent bed surface is able to resist all flows but those whose largest value of the fluctuating fluid shear stress τ^{fluc} , associated with the most energetic turbulent eddy that can possibly form (note that the size, and thus the energy content, of eddies is limited by the system dimensions, Smits et al., 2011), exceeds a certain critical resisting shear stress $\Theta_Y \rho_p \tilde{g} d$ (Clark et al., 2017). In particular, for laminar

where the overbar denotes the particle concentration (ρ)-weighted height average, $\bar{\cdot} \equiv \frac{1}{M} \int_0^{z_{\max}} \rho \langle \cdot \rangle dz$, with $z_{\max}$ the top of the transport layer and $M \equiv \int_0^{z_{\max}} \rho dz$ (the same M as $M = \rho_p d M^*$ in Equation 2b). The grain acceleration $\mathbf{a}$ consists of a drag (superscript d), a gravity (superscript g), and a buoyancy (superscript b) component. In particular, $a_z = a_z^d + a_z^g + a_z^b = a_z^d - \tilde{g}$. Hence, defining $S \equiv -(a_x^g + a_x^b) / (a_z^g + a_z^b)$, $\mathbf{a}$ can be expressed as

$$(a_x, a_z) = (a_x^d + S\tilde{g}, a_z^d - \tilde{g}). \quad (5)$$

Note that $S \simeq (1 - 1/s)^{-1} \tan \alpha$ for slope-driven NST in turbulent liquids and $S \simeq \tan \alpha$ for aeolian NST and slope-driven NST in viscous liquids (these differences arise because $\mathbf{a}^b$ is proportional to the divergence of only the viscous contribution to the fluid stress tensor, Maurin et al., 2018). Since these conditions cover most natural environments, S is treated as a further constant dimensionless number (the *slope number*) characterizing a given NST threshold condition in addition to Ga and s .

In order to allow for an easy analytical evaluation of Equation 4, we linearize $\mathbf{a}^d$ via approximating the difference $|\mathbf{u} - \mathbf{v}|$ between fluid ($\mathbf{u}$) and grain ($\mathbf{v}$) velocity by the mean value of its streamwise component: $|\mathbf{u} - \mathbf{v}| \approx \bar{u}_x - \bar{v}_x$. We carried out a few tests with the final transport threshold model that suggested that this approximation has almost no effect on the final prediction. Using a standard drag law for spherical grains (Camenen, 2007; Ferguson & Church, 2004), the linearized drag acceleration reads

$$\mathbf{a}^d = \frac{3}{4sd} \left[m^d \sqrt{\frac{24\sqrt{s\tilde{g}d}}{Ga}} + m^d \sqrt{C_d^\infty (\bar{u}_x - \bar{v}_x)} \right] (\mathbf{u} - \mathbf{v}), \quad (6)$$

where $C_d^\infty = 0.4$ and $m^d = 2$ (for nonspherical grains, different values of C_d^∞ and m^d are more appropriate, Camenen, 2007). Equation 6 implies two mathematical identities. First, $a_x^d / \tilde{g} = (\bar{u}_x - \bar{v}_x) / v_s$, where $v_s \equiv -v_z|_{a_z^d = \tilde{g}}$ is the terminal grain settling velocity, because of $a_x^d / a_z^d = (u_x - v_x) / (u_z - v_z)$ and $u_z = 0$. Second, $a_z^d = 0$ because of $\bar{v}_z = 0$ (mass conservation, Pahltz et al., 2015) and $u_z = 0$. Using Equations 4 and 5, $a_z^d = 0$ further implies $a_x^d / \tilde{g} = \mu_b - S$. After using the latter relation in Equation 6 and taking the m^d -th root, it follows that $[(\bar{u}_x - \bar{v}_x) / \sqrt{s\tilde{g}d}]^{1/m^d}$ obeys a quadratic equation with coefficients that depend on $\mu_b - S$ and Ga . Solving this quadratic equation and combining it with the first mathematical identity, we obtain the following expression for $\bar{u}_x - \bar{v}_x$ and the dimensional (nondimensionalized) settling velocity v_s (v_{s^*}):

$$\begin{aligned} v_{s^*} &\equiv \frac{v_s}{\sqrt{s\tilde{g}d}} = \frac{\bar{u}_x - \bar{v}_x}{(\mu_b - S)\sqrt{s\tilde{g}d}} \\ &= \frac{1}{\mu_b - S} \left[\sqrt{\frac{1}{4} m^d \sqrt{\left(\frac{24}{C_d^\infty Ga} \right)^2}} + m^d \sqrt{\frac{4(\mu_b - S)}{3C_d^\infty}} - \frac{1}{2} m^d \sqrt{\frac{24}{C_d^\infty Ga}} \right]^{m^d}. \end{aligned} \quad (7)$$

A similar link between $\bar{u}_x - \bar{v}_x$ and v_s as in Equation 7 was previously established by Bagnold (1973), while the expression for $\bar{u}_x - \bar{v}_x$ has been validated with data from DEM-based numerical simulations of NST for a wide range of conditions (Pahltz & Duran, 2018a). Both facts support using a linearized drag law (Equation 6). Note that the terminal grain settling velocity in quiescent flow v_s^- , which is distinct from v_s , obeys a modified version of the lower line of Equation 7 in which $\mu_b - S$ is replaced by 1 (Camenen, 2007).

Equation 7, when combined with insights from previous DEM-based numerical simulations of NST into the physical nature of μ_b , supports representing the entire grain motion in equilibrium NST by grains saltating in identical periodic trajectories with rebound boundary conditions (Appendix C).

2.3. Mathematical Description of Periodic Saltation

This subsection introduces the mathematical description of the main model idealization: saltation in identical periodic trajectories along a flat wall. Equation 7 is a part of this description. However, it is not immediately clear what are the physical meanings of μ_b and the concentration (ρ)-weighted height average $\bar{\cdot}$,

both appearing in Equation 7, in the context of periodic saltation. Below, we therefore derive expressions for both items in terms of three quantities that are intuitively associated with periodic saltation: the lift-off or rebound velocity $\mathbf{v}_r$, impact velocity $\mathbf{v}_i$, and hop time T .

In periodic saltation, a grain crosses each elevation $z < H$, where H is the hop height, twice during a single saltation trajectory: once for times before (superscript $\uparrow$) and once for times after (superscript $\downarrow$) the instant $t(H)$ at which it reaches its highest elevation $z = H$. In particular, due to mass conservation, the upward vertical mass flux of grains $\phi^\uparrow$, which is equal to the negative downward vertical mass flux of grains, is constant for $z < H$, while $\phi^\uparrow = 0$ for $z > H$ (Berzi et al., 2016). Hence, for $z < H$, the concentrations of ascending and descending grains are given by $\rho^\uparrow(z) = \phi^\uparrow / v_z^\uparrow(z)$ and $\rho^\downarrow(z) = -\phi^\uparrow / v_z^\downarrow(z)$, respectively (note that $\mathbf{v}^{\uparrow(\downarrow)}(0) = \mathbf{v}_r^{\uparrow(\downarrow)}$). Using $\rho(z) = \rho^\uparrow(z) + \rho^\downarrow(z)$, it follows that the ρ -weighted height average of a grain quantity G is equal to its time average:

$$\bar{G} = \frac{\int_0^{z_{\max}} \rho \langle G \rangle dz}{\int_0^{z_{\max}} \rho dz} = \frac{\phi^\uparrow \int_0^H \left(\frac{G^\uparrow}{v_z^\uparrow} - \frac{G^\downarrow}{v_z^\downarrow} \right) dz}{\phi^\uparrow \int_0^H \left(\frac{1}{v_z^\uparrow} - \frac{1}{v_z^\downarrow} \right) dz} = \frac{\int_0^{t(H)} G^\uparrow dt - \int_T^{t(H)} G^\downarrow dt}{\int_0^{t(H)} dt - \int_T^{t(H)} dt} = \frac{1}{T} \int_0^T G dt. \quad (8)$$

Furthermore, using $\mu_b = -\langle v_z' v_x' \rangle(0) / \langle v_z'^2 \rangle(0)$ (Appendix C) and $\langle v_z' v_x' \rangle = \langle v_z v_i \rangle$ because of $\langle v_z \rangle = 0$ (mass conservation, Pähtz et al., 2015), it also follows that one can link μ_b to the rebound and impact velocities:

$$\mu_b = -\frac{[\rho^\uparrow v_z^\uparrow v_x^\uparrow](0) + [\rho^\downarrow v_z^\downarrow v_x^\downarrow](0)}{[\rho^\uparrow v_z^{\uparrow 2}](0) + [\rho^\downarrow v_z^{\downarrow 2}](0)} = \frac{v_{\downarrow x} - v_{\uparrow x}}{v_{\uparrow z} - v_{\downarrow z}}. \quad (9)$$

Now, we subdivide this subsection into further subsections. First, we present the deterministic laws governing the motion of a grain above the bed driven by the mean turbulent flow (Section 2.3.1). These laws directly map $\mathbf{v}_r$ to $\mathbf{v}_i$. Second, we present the laws describing grain-bed rebounds (Section 2.3.2), mapping $\mathbf{v}_i$ back to $\mathbf{v}_r$. For the grain trajectories to be identical and periodic, these laws must also be deterministic, which is achieved by representing them by their statistical mean effect. Third, we model the critical Shields number Θ^{roll} needed for a grain of a given kinetic energy, associated with a given periodic saltation trajectory, to roll (or hop) out of the most stable pockets of the bed surface assisted by the near-surface flow (Section 2.3.3). For a periodic saltation trajectory to be consistent with a sustained rolling motion, we require that $\Theta > \Theta^{\text{roll}}$.

2.3.1. Grain Motion Above the Bed

To make the analytical notation compact, we nondimensionalize location, velocity, acceleration, and time, indicated by a hat, using combinations of the terminal settling velocity and reduced gravity: $v_s^2 / \tilde{g}$, v_s , $\tilde{g}$, and $v_s / \tilde{g}$, respectively. Using $\hat{z} \equiv z / (v_s^2 / \tilde{g}) = z / (v_s^2 s d)$, one then obtains the following system of differential equations describing the average trajectory from Equations 3, 5, and 6:

$$\hat{u}_x(\hat{z}) = v_s^{-1} \sqrt{\Theta} f_u(Ga \sqrt{\Theta}, s v_s^2 \hat{z}), \quad (10a)$$

$$\frac{d}{d\hat{t}} \hat{v}_x = \hat{u}_x - \hat{v}_x + S, \quad (10b)$$

$$\frac{d}{d\hat{t}} \hat{v}_z = -1 - \hat{v}_z. \quad (10c)$$

The solution of Equations 10a–10c, with the initial condition $\hat{\mathbf{v}}(0) = \hat{\mathbf{v}}_\uparrow$, is straightforward and given in Appendix E. For the transport threshold model, the following expressions, which can be obtained from the solution (Appendix E), are crucial (written in a form that allows easy iterative evaluation, see Section 2.4):

$$\hat{v}_{\downarrow z} = -1 - W \left[-(1 + \hat{v}_{\uparrow z}) e^{-(1 + \hat{v}_{\uparrow z})} \right], \quad (11)$$

$$\Theta = \frac{\sqrt{\Theta} v_{s*} [\mu_b (1 + \hat{v}_{\uparrow z}) + \hat{v}_{\uparrow x} - S]}{f_u \left\{ Ga \sqrt{\Theta} v_{s*}^2 [-\hat{v}_{\downarrow z} (1 + \hat{v}_{\uparrow z}) - \hat{v}_{\uparrow z}] \right\}}, \quad (12)$$

where W denotes the principal branch of the Lambert- W function.

2.3.2. Grain-Bed Rebounds

Grain collisions with a static sediment bed have been extensively studied experimentally (Ammi et al., 2009; Beladjine et al., 2007), numerically (Comola, Gaume, et al., 2019; Lämmel et al., 2017; Tanabe et al., 2017), and analytically (Comola & Lehning, 2017; Lämmel et al., 2017). In typical experiments, an incident grain is shot with a relatively high impact velocity ($|\mathbf{v}_{\downarrow}| \gg \sqrt{gd}$) onto the bed and the outcome of this impact (i.e., the grain rebound and potentially ejected bed grains) statistically analyzed. We describe this process using a phenomenological description for the average rebound (vertical) restitution coefficient $e \equiv |\mathbf{v}_{\uparrow}|/|\mathbf{v}_{\downarrow}|$ ($e_z \equiv -v_{\uparrow z}/v_{\downarrow z}$) as a function of $\sin \theta_{\downarrow} = -v_{\downarrow z}/|\mathbf{v}_{\downarrow}|$, with $\theta_{\downarrow}$ the average impact angle (Beladjine et al., 2007):

$$e = A - B \sin \theta_{\downarrow}, \quad (13a)$$

$$\text{Original: } e_z = A_z / \sin \theta_{\downarrow} - B_z, \quad (13b)$$

$$\text{Modified: } e_z = (A + C) / \sqrt{\sin \theta_{\downarrow}} - (B + C), \quad (13c)$$

where $A = 0.87$, $B = 0.72$, $A_z = 0.3$, $B_z = 0.15$, and $C = 0$. Equation 13c is our modification of Equation 13b, the original expression given by Beladjine et al. (2007). This modification accounts for the analytically derived asymptotic behavior of the rebound angle in the limit of small impact angle, $\sin \theta_{\uparrow} = v_{\uparrow z} / |\mathbf{v}_{\uparrow}| = e_z \sin \theta_{\downarrow} / e \sim \sqrt{\sin \theta_{\downarrow}}$ (Lämmel et al., 2017), and for the requirement that $\theta_{\uparrow} \rightarrow 90^\circ$ when $\theta_{\downarrow} \rightarrow 90^\circ$. Like the original expressions, the modified expressions are consistent with experimental data by Beladjine et al. (2007) for nearly monodisperse, cohesionless, spherical grains, as shown in Figures 2a–2c.

We assume that Equations 13a and 13c are roughly universal for monodisperse, cohesionless, spherical grains, independent of $|\mathbf{v}_{\downarrow}|$ and bed-related parameters, such as ρ_p , $\tilde{g}$, and d , since the experimental data by Beladjine et al. (2007) have also been reproduced by a theoretical model that predicts the rebound parameters as a function of only $\theta_{\downarrow}$ (Lämmel et al., 2017). Furthermore, for conditions in which vertical drag is negligible (i.e., $|a_z^d| \ll \tilde{g}$, $e_z \simeq 1$, $\overline{v_x} \simeq (v_{\uparrow x} + v_{\downarrow x}) / 2$, and $\overline{v_z^2} \simeq v_{\uparrow z}^2 / 3$, see Appendix F), any given set of rebound laws that depends only on $\theta_{\downarrow}$, such as Equations 13a and 13c, results in $v_{\uparrow z} = -v_{\downarrow z} \propto v_{\uparrow x} \propto v_{\downarrow x}$ with fixed proportionality constants, implying $\mu_b = \text{const}$ (Equation 9) and $(\overline{v_z^2})^{1/2} \propto \overline{v_x}$. Qualitatively, these two scaling relations correspond to the semiempirical closures used in the threshold model of Pärtz and Durán (2018a) (equations C2a and C2b), based on which we motivate the usage of universal rebound boundary conditions across all NST regimes in Appendix C.

Figures 2d and 2e show that, for the DEM numerical simulation data by Comola, Gaume, et al. (2019), e and e_z are insensitive to the cohesiveness of the bed material. In fact, for a bed consisting of spherical grains with log-normally distributed size and a grain impacting with a relatively high-impact velocity ($|\mathbf{v}_{\downarrow}| \gg \sqrt{gd}$), these authors varied the critical force F_ϕ that is required to break cohesive bonds between bed grains over several orders of magnitude and found nearly no effect on the average rebound dynamics. This finding will play a crucial role in Section 4.2, where we discuss the importance of cohesion for NST.

Lastly, we note that viscous damping of binary collisions (e.g., Gondret et al., 2002), which can be important for fluvial NST, also does not seem to significantly affect the rebound laws (Pärtz et al., 2020, Section 4.1.1.4), in contrast to the assumptions in previous trajectory-based transport threshold models (Berzi et al., 2016, 2017). In particular, DEM-based simulations of aeolian and fluvial NST indicate that, even for nearly fully damped binary collisions (normal restitution coefficient $\epsilon = 0.01$), grain-bed rebounds (and thus e and e_z) are not much affected when compared with nondamped ($\epsilon = 0.9$) binary collisions (Pärtz & Durán, 2018a, see particularly their Movies S1–S3). This probably means that the tangential relative velocity component, which is not much affected by ϵ , dominates during the rebound process.

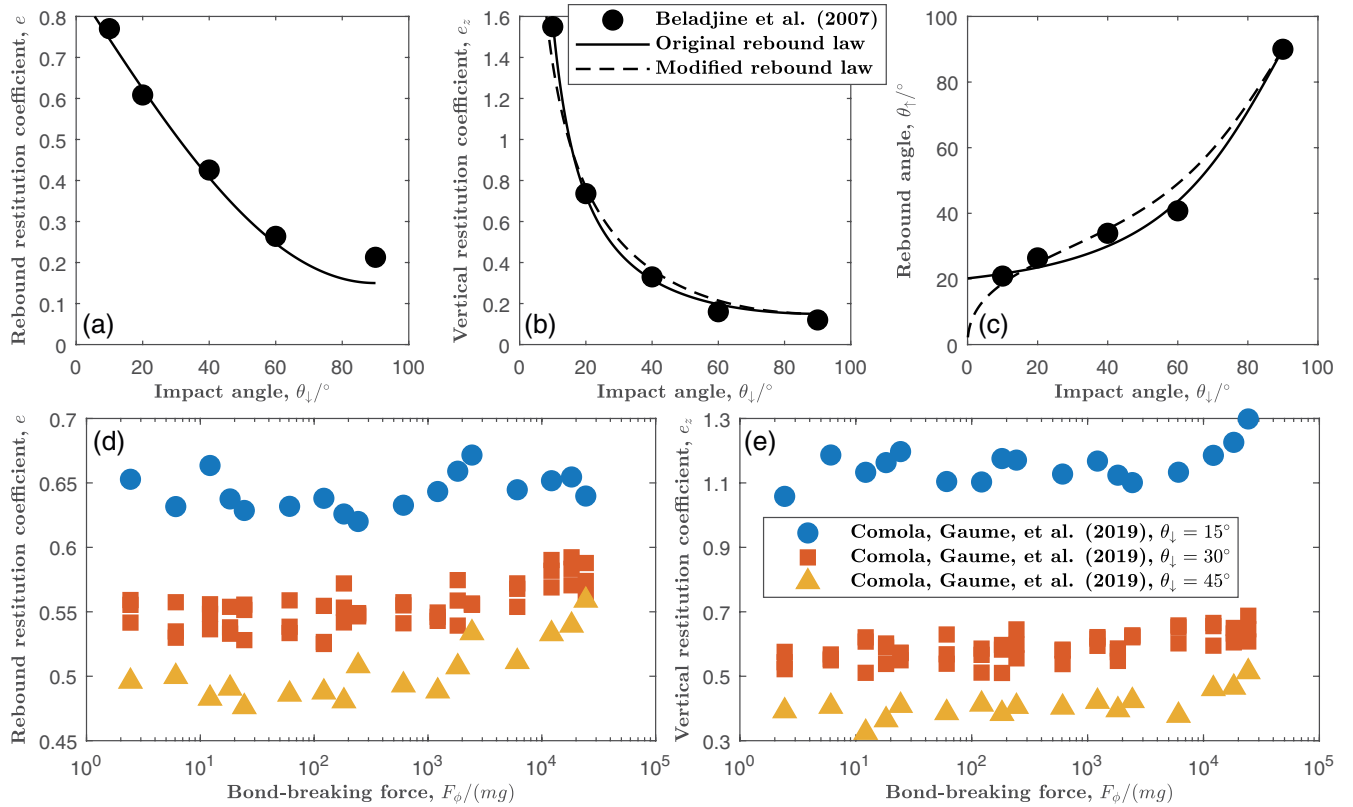


Figure 2. Rebound laws. (a) Rebound restitution coefficient e , (b) vertical restitution coefficient e_z , and (c) rebound angle $\theta_r = \arcsin(e_z \sin \theta_i / e)$ versus impact angle θ_i . (d) Rebound restitution coefficient e and (e) vertical restitution coefficient e_z versus the dimensionless force $F_\phi/(mg)$ needed to break cohesive bonds between bed grains. Symbols in (a)–(c) correspond to averaged experimental data for nearly monodisperse, cohesionless, spherical grains ($\rho_p = 1770 \text{ kg/m}^3$ and $d = 6 \text{ mm}$) (Beladjine et al., 2007). The solid line in (a) corresponds to Equation 13a. The solid (dashed) lines in (b) and (c) correspond to Equation 13b (Equation 13c). Symbols in (d) and (e) correspond to averaged DEM numerical simulation data by Comola, Gaume, et al. (2019) for $\rho_p = 1,000 \text{ kg/m}^3$, $g = 9.81 \text{ m/s}^2$, $m = \rho_p \pi d^3 / 6$, three different impact velocities $|\mathbf{v}_i| = (1.58, 3.16, 4.74) \text{ m/s}$, and three different impact angles $\theta_i = (15^\circ, 30^\circ, 45^\circ)$. The rebounding spherical grain has a diameter of $d = 200 \mu\text{m}$, while the diameter of the spherical bed grains obeys a log-normal distribution with mean $\bar{d} = 200 \mu\text{m}$ and standard deviation $\sigma_d = 50 \mu\text{m}$. The figure legend in (d) is the same as in (e). Figure legends in (a) and (c) are the same as in (b). DEM, discrete element method.

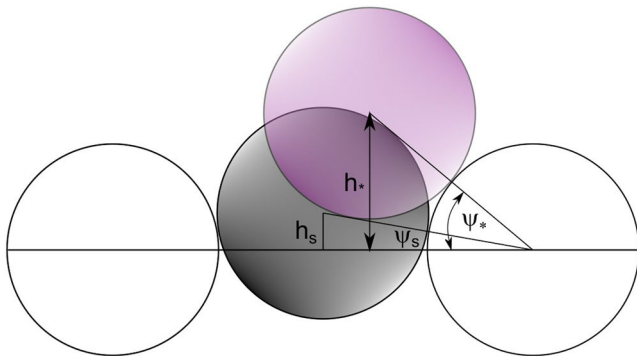


Figure 3. Sketch of a grain of a given kinetic energy E_γ rolling out of the most stable bed surface pocket. The process is modeled as follows. The grain first rolls along its downstream neighbor (neglecting rolling friction and flow driving) until E_γ has been fully converted into potential energy $m\tilde{g}(h_* - h_s)$, increasing the pocket angle via $\sin \psi_* = \sin \psi_s + E_\gamma / (m\tilde{g}d)$. A sufficiently strong flow can then push the grain out from this new location within the pocket.

2.3.3. Critical Shields Number Required for Rolling

In Appendix C, we provide justifications for why one can represent the entire grain motion in NST, including grains that roll and/or slide along the bed, by a pure saltation motion. However, to be consistent with a sustained rolling motion of grains, we require that saltation trajectories are limited to Shields numbers Θ that are larger than the critical value Θ^{roll} needed for a grain of a given kinetic energy, associated with a given periodic saltation trajectory, to roll (or hop) out of the most stable pockets of the bed surface assisted by the near-surface flow. In this section, we derive an expression for Θ^{roll} using a highly simplified approach. First, since a grain located in such a pocket just changed its direction of motion from downward to upward, we assume that it exhibits the rebound kinetic energy $E_\gamma = \frac{1}{2} m \mathbf{v}_\gamma^2$, where m is the grain mass, of the given periodic saltation trajectory. Second, we assume that this grain first rolls along its downstream neighbor until E_γ has been fully converted into potential energy $m\tilde{g}(h_* - h_s)$ (Figure 3), neglecting rolling friction and flow driving.

This rolling motion increases the pocket angle from the value ψ_s corresponding to the most stable bed surface pocket to the value ψ_* via (Figure 3)

$$\sin \psi_* = \sin \psi_s + E_{\uparrow} / (m\tilde{g}d). \quad (14)$$

We then model Θ^{roll} as the critical Shields number required to push the grain out from this new pocket angle position, assuming that the mean grain motion driven by a turbulent flow is the same as the mean grain motion driven by the mean turbulent flow (the first idealization in Section 2).

Our definition of Θ^{roll} implies that Θ^{roll} depends on $E_{\uparrow} / (m\tilde{g}d)$ via Equation 14 and vanishes for sufficiently large values of $E_{\uparrow} / (m\tilde{g}d)$, for which the grain can roll (or hop) out of the pocket all by itself (i.e., for $\psi_* \geq 90^\circ$). Furthermore, in the limit $E_{\uparrow} / (m\tilde{g}d) \rightarrow 0$ (i.e., $\psi_* \rightarrow \psi_s$), $\Theta^{\text{roll}} \rightarrow \Theta_Y$, since the yield stress Θ_Y can be interpreted as the Shields number required to initiate rolling of grains resting in the most stable pockets of the bed surface in the absence of turbulent fluctuations (Section 2.1.3). This limit is relevant for $Ga^2s \lesssim 1$, typical for NST driven by laminar fluvial flows (Pähtz & Durán, 2018a), where the average energy of transported grains scales with $Ga^2sm\tilde{g}d$ (Charru et al., 2004) and thus $E_{\uparrow} / (m\tilde{g}d)$ is relatively small.

For a nonsloped ($S = 0$) bed of triangular or quadratic geometry and a laminar driving flow, Agudo et al. (2017) derived nearly exact analytical expressions for the critical Shields number Θ_c required to push a grain out from an arbitrary pocket angle position ψ . As triangular arrangements are the most probable ones in disordered configurations (Agudo et al., 2017), we assume that these expressions approximately apply also to natural sediment beds. For the special case $\psi = 25^\circ$, these expressions predict $\Theta_{c|\psi=25^\circ} \simeq 0.13$. Since the nonsloped yield stress exhibits the same value, $\Theta_Y^o = 0.13$ (Section 2.1.3), we identify this pocket angle as the one of the most stable bed surface pocket (i.e., $\psi_s = 25^\circ$). Agudo et al. (2017) further noted that their analytical expressions are reasonably well approximated by the nonsloped version of the model of Wiberg and Smith (1987): $\Theta_c \propto \cot \psi$ for $S = 0$. Assuming that the general version of the model of Wiberg and Smith (1987), $\Theta_c \propto \cot \psi - S$, works reasonably well for arbitrarily sloped beds (which have not been studied by Agudo et al., 2017), we conclude that $\Theta_c = (\cot \psi - S)\Theta_Y^o / \cot \psi_s$. Hence, identifying Θ^{roll} as $\Theta_c|_{\psi=\psi_*}$, using Equation 14 with $E_{\uparrow} / (m\tilde{g}d) = \frac{1}{2}sv_{s*}^2\hat{\mathbf{v}}_{\uparrow}^2$, and imposing that $\Theta^{\text{roll}} = 0$ when the grain exhibits a sufficient kinetic energy to be able to roll (or hop) out the most stable bed surface pocket all by itself, we obtain

$$\Theta^{\text{roll}} = \frac{\Theta_Y^o}{\cot \psi_s} \max[\cot \psi_* - S \text{sgn}(\cot \psi_*), 0], \text{ with} \quad (15)$$

$$\cot \psi_* = \sqrt{\max\left[\left(\sin \psi_s + \frac{1}{2}sv_{s*}^2\hat{\mathbf{v}}_{\uparrow}^2\right)^{-2} - 1, 0\right]},$$

where sgn denotes the sign function (note that $\text{sgn}(0) = 0$). Note that $\Theta^{\text{roll}} = 0$ does not imply that a flow with Shields number $\Theta = 0$ is able to sustain a rolling motion of grains because Θ^{roll} depends on the value of $\mathbf{v}_{\uparrow}$, which is associated with a given periodic saltation trajectory, and for $\Theta = 0$, such a trajectory does not exist in the first place.

2.4. Computation of Threshold and Rate of Equilibrium NST

From solving Equations 3, 7, 9, 11, 12, 13a, and 13c, we obtain a family of identical periodic trajectory solutions $\Theta(Ga, s, S, \hat{\mathbf{v}}_{\uparrow z})$. In detail, for given values of Ga , s , S , and $\hat{\mathbf{v}}_{\uparrow z}$, Θ is obtained in the following manner:

1. Compute $\hat{\mathbf{v}}_{\downarrow z}$ using Equation 11
2. Compute $\hat{\mathbf{v}}_{\uparrow x}$ and $\hat{\mathbf{v}}_{\downarrow x}$ using Equations 13a and 13c
3. Compute v_{s*} and μ_b using Equation 7 and Equation 9, respectively
4. Iteratively solve Equation 12 for Θ using Equation 3. An initial value that works well is $\Theta = 1$.

Every periodic trajectory solution $\Theta(Ga, s, S, \hat{\mathbf{v}}_{\uparrow z})$ exhibits a certain value of $sv_{s*}^2\hat{\mathbf{v}}_{\uparrow}^2$. Hence, Θ^{roll} , calculated as a function of $sv_{s*}^2\hat{\mathbf{v}}_{\uparrow}^2$ (Equation 15), can also be parametrized by Ga , s , S , and $\hat{\mathbf{v}}_{\uparrow z}$. Then, from those periodic

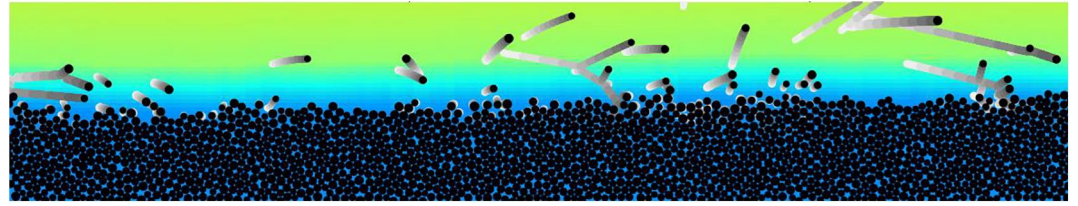


Figure 4. Snapshot of a simulation of NST using the DEM-based model of Durán et al. (2012) showing a quarter of the longitudinal simulation domain. This model couples a DEM for the particle motion under gravity, buoyancy, and fluid drag with a continuum Reynolds-averaged description of hydrodynamics. Spherical particles ($\sim 10^4$) with mild polydispersity are confined in a quasi-two-dimensional domain of length $\sim 10^3 d$, with periodic boundary conditions in the flow direction, and interact via normal repulsion (restitution coefficient $\epsilon = 0.9$) and tangential friction (contact friction coefficient $\mu_c = 0.5$). The bottom-most particle layer is glued on a bottom wall, while the top of the simulation domain is reflective but so high that it is never reached by transported particles. The Reynolds-averaged Navier-Stokes equations are combined with a semiempirical mixing length closure that ensures a smooth hydrodynamic transition from high to low particle concentration at the bed surface and quantitatively reproduces the mean turbulent flow velocity profile in the absence of transport. DEM, discrete element method; NST, nonsuspended sediment transport.

trajectory solutions that satisfy $\Theta(Ga, s, S, \hat{v}_{\uparrow z}) \geq \Theta^{\text{roll}}(Ga, s, S, \hat{v}_{\uparrow z})$, we obtain the transport threshold from minimizing Θ as a function of $\hat{v}_{\uparrow z}$:

$$\Theta_t(Ga, s, S) \equiv \min_{\hat{v}_{\uparrow z}: \Theta(Ga, s, S, \hat{v}_{\uparrow z}) \geq \Theta^{\text{roll}}(Ga, s, S, \hat{v}_{\uparrow z})} \Theta(Ga, s, S, \hat{v}_{\uparrow z}). \quad (16)$$

Note that, for conditions corresponding to the *bedload* regime, associated with low-energy threshold trajectories (a precise definition is provided in Section 3.2), periodic trajectory solutions $\Theta(Ga, s, S, \hat{v}_{\uparrow z})$ tend to be monotonously increasing with $\hat{v}_{\uparrow z}$, while $\Theta^{\text{roll}}(Ga, s, S, \hat{v}_{\uparrow z})$ is always monotonously decreasing with $\hat{v}_{\uparrow z}$ (Equation 15). That is, the threshold trajectory calculated from Equation 16 tends to satisfy $\Theta_t = \Theta_t^{\text{roll}} \equiv \Theta^{\text{roll}}|_{\Theta=\Theta_t}$ for such conditions.

From the threshold trajectory, we obtain the threshold bed friction coefficient μ_{bt} and dimensionless average streamwise grain velocity $\overline{v_{x*}}_t$ using Equations 9, 10a, E11a, and E11b:

$$\mu_{bt} = (v_{\downarrow xt} - v_{\uparrow xt}) / (v_{\uparrow xt} - v_{\downarrow xt}), \quad (17)$$

$$\overline{v_{x*}}_t = \sqrt{\Theta_t} f_u \left[Ga \sqrt{\Theta_t}, \frac{1}{2} s v_{s*}^2 (\hat{v}_{\uparrow zt} + \hat{v}_{\downarrow zt}) \right] + (S - \mu_{bt}) v_{s*}. \quad (18)$$

Lastly, from μ_{bt} , $\overline{v_{x*}}_t$, Θ_t , and $c_M = 1.7$, we calculate the dimensionless rate Q_* of equilibrium NST via Equations 2a and 2b using μ_{bt} as the value of μ_b in Equation 2b, since DEM-based numerical simulations of NST indicate that μ_b does not significantly change with Θ (Pähtz & Durán, 2018b).

3. Results

3.1. Model Evaluation With Experimental and Numerical Data

This section compares the model predictions with NST data from many experimental studies and with data from DEM-based numerical simulations of NST by Pähtz and Durán (2018a); Pähtz and Durán (2020) using the numerical model of Durán et al. (2012) (a snapshot and brief description are provided in Figure 4). We start with the comparison to the numerical data in order to explore the range of validity of the model. To this end, the form drag coefficient in Equation 7 is modified to the value $C_d^\infty = 0.5$ used by Pähtz and Durán (2018a); Pähtz and Durán (2020). Furthermore, since Pähtz and Durán (2018a); Pähtz and Durán (2020) simulated a quasi-two-dimensional system, parameters characterizing the bed surface also need to be modified. We did so manually and found that the following modified values lead to good overall agreement with the numerical data: $\Theta_Y^o = 0.18$, $A = 0.8$, $B = 0.3$, and $C = 0.3$. In fact, Figure 5a shows that the modified model is consistent with the simulated transport thresholds, calculated via extrapolating $(\mu_b - \tan \alpha)M_*$ to zero (consistent with the definition of Θ_t in Equation 2b), across a large range of the

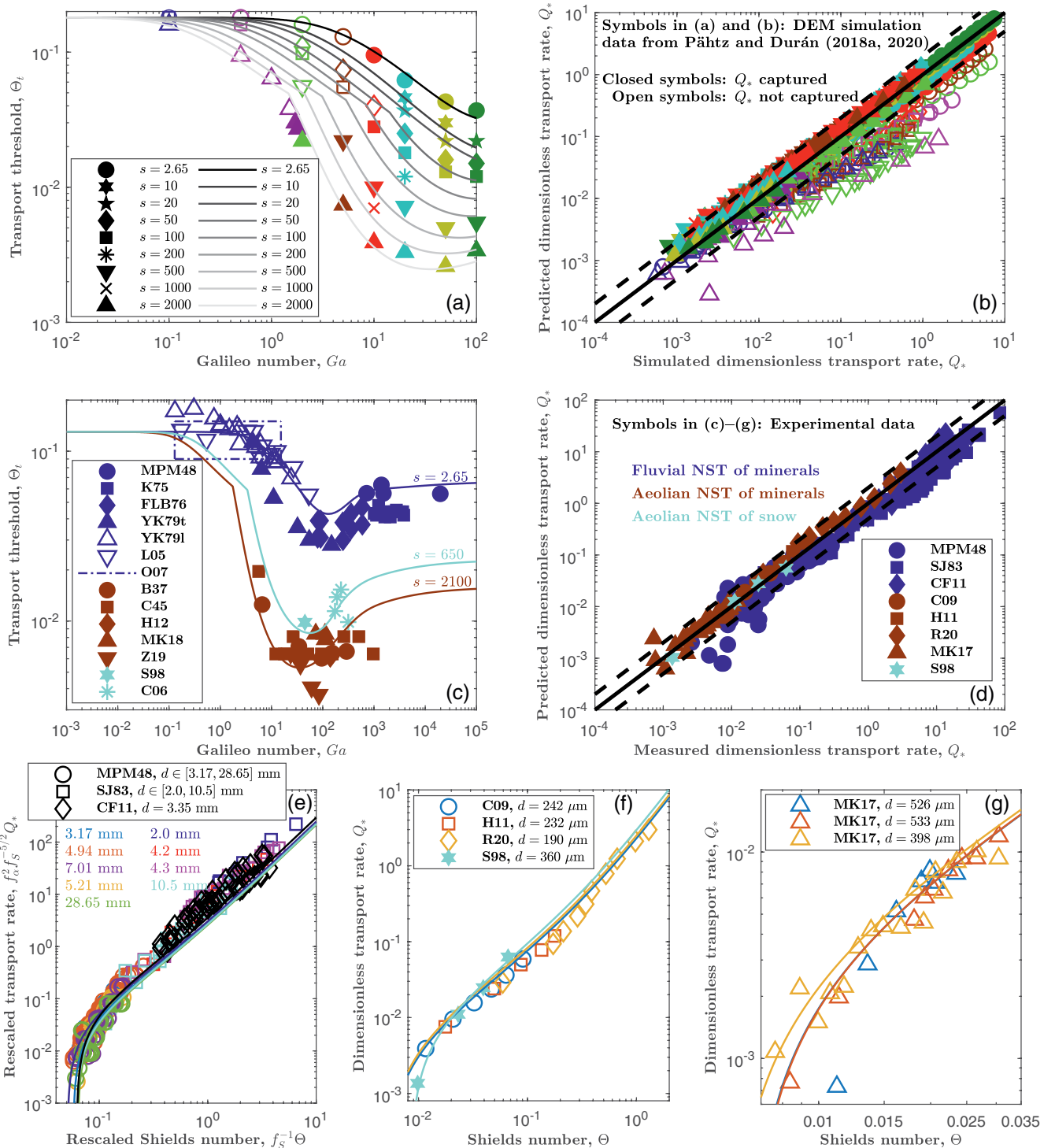


Figure 5. Model evaluation with experimental and numerical data. (a) and (c) Transport threshold Θ_t versus Galileo number Ga for varying density ratio s . (b) and (d) Predicted versus simulated (b) or measured (d) dimensionless sediment transport rate Q_s . (e)–(g) Rescaled sediment transport rate $f_\alpha^2 f_s^{-5/2} Q_s$ versus rescaled Shields number $f_s^{-1} \Theta$, where the slope correction factors $f_\alpha = f_s = 1$ in (f) and (g) as $\alpha = 0$. Symbols in (a) and (b) correspond to data from DEM-based numerical simulations of NST by Pätz and Durán (2018a); Pätz and Durán (2020) for $S = 0$ and varying Ga and s , as indicated in the legend and plot in (a), and varying Θ (b). Symbols in (c)–(g) correspond to experimental data of various studies (Sections 3.1.1–3.1.4). Experimental data by Ouriemi et al. (2007), who did not report single data points, lie within the dash-dotted rectangle in (c). Solid lines in (c), (d), (f), and (g) correspond to model predictions. Solid lines in (e) correspond to predictions of the model in which Equations 2a and 2b have been approximated by Equations 20a and 20b, respectively. Solid lines in (a) and (b) correspond to model predictions using modified parameter values adjusted to the simulations (Section 3.1). Dashed lines in (b) and (d) indicate a deviation from the predictions by a factor of 2. DEM, discrete element method; NST, nonsuspended sediment transport.

Galileo number and density ratio: $Ga \in [0.1, 100]$ and $s \in [2.65, 2000]$. Furthermore, Figure 5b shows that the modified model also captures the simulated transport rate data for conditions that satisfy

$$Ga\sqrt{s} \gtrsim \begin{cases} 15, & s < 10 \\ 75, & s \geq 10 \end{cases} \quad (19)$$

within a factor of 2 (closed symbols), whereas conditions that do not satisfy this constraint are not captured (open symbols). The reason behind this restriction in the model's validity range is that $s^{1/2} Ga$ is a Stokes-like number that encodes the importance of grain inertia relative to viscous drag forcing (Pähtz & Durán, 2017; Pähtz & Durán, 2018a). In fact, the validity of Equation 2a requires that, within the dense, highly collisional part of the transport layer, the motion of a transported grain is not much affected by fluid drag between two subsequent collisions with other transported grains (Pähtz & Durán, 2020). Note that collisions between transported grains are not relevant for the transport threshold model, which is why its validity is not affected by $s^{1/2} Ga$ (Figure 5a).

Figures 5c–5g show that the nonmodified model simultaneously captures several experimental data sets corresponding to fluvial and aeolian NST within a factor of about 2. Note that all the threshold data chosen for the model evaluation were measured using methods that are known or suspected to be consistent with the definition of Θ_t by Equation 2b. In the following subsection, the experimental data sets are described in detail.

3.1.1. Fluvial Transport Threshold Data Sets

For NST driven by laminar flows, flume measurements of the visual initiation threshold by Yalin and Karahan (1979, YK79l, $s \in (1.9, 2.3)$) and Loiseleux et al. (2005, L05, $s \in (2.1, 2.5)$) and of the visual cessation threshold by Ouriemi et al. (2007, O07, $s \in (1.0, 2.5)$) are shown (open symbols and dash-dotted lines in Figure 5c). Note that, for such conditions, both thresholds are approximately equivalent to each other and Θ_t (Clark et al., 2017; Pähtz & Durán, 2018a). Furthermore, flume measurements of the visual initiation threshold for subaqueous bedload by Karahan (1975, K75, $s \simeq 2.65$), Fernandez Luque and van Beek (1976, FLB76, $s \in (1.3, 4.6)$), and Yalin and Karahan (1979, YK79t, $s \simeq 2.65$) are shown (closed symbols in Figure 5c), where critical conditions are defined by a critical value of a proxy characterizing the state of transport, such as a critical value of the nondimensionalized bed sediment entrainment rate (Paphitis, 2001, Equation 7). It has been argued that the resulting threshold is close to Θ_t because such and similar critical transport conditions are associated with an abrupt sharp transition in the behavior of Q_* as a function of Θ (Pähtz et al., 2020). Furthermore, we obtained Θ_t from extrapolating the paired flume measurements of Q_* and Θ by Meyer-Peter and Müller (1948, MPM48, those in Figure 5e, $s \in (2.6, 2.7)$) to vanishing Q_* using the function $Q_* = c_1(\Theta - \Theta_t) + c_2(\Theta - \Theta_t)^2$ (consistent with Equations 2a and 2b, Pähtz & Durán, 2020), where we treated c_1 , c_2 , and Θ_t as fit parameters.

3.1.2. Aeolian Transport Threshold Data Sets

For aeolian NST, we consider data that have been obtained from extrapolating equilibrium transport conditions to vanishing transport, consistent with the definition of Θ_t in Equation 2b as the value at which the equilibrium value of M_* vanishes. Following previous studies (Comola, Kok, et al., 2019; Martin & Kok, 2018; Pähtz et al., 2020), we also assume that the cessation threshold of intermittent saltation is identical to Θ_t . However, we do not consider measurements of the continuous transport threshold (discussed in Section 4.1) and of the threshold of saltation initiation.

Wind tunnel measurements for windblown sand from Ho (2012, H12, $s \simeq 2060$) and Zhu et al. (2019, Z19, $s \simeq 2210$) are shown in Figure 5c, who carried out an indirect extrapolation to vanishing transport to obtain Θ_t using a proxy of Q_* : the surface roughness z_0 , which undergoes a regime shift when transport ceases. Furthermore, visual wind tunnel measurements for windblown sand and snow by Bagnold (1937, B37, $s \simeq 2210$), Chepil (1945, C45, $s \in (1,400, 1800)$), and Sugiura et al. (1998, S98, $s \simeq 650$) are shown, who obtained the cessation threshold of intermittent saltation (Θ_t) as the smallest value of Θ for which energetic mineral or snow grains fed at the tunnel entrance are able to saltate along the bed without stopping. Direct field measurements of this threshold based on the *Time Frequency Equivalence Method* (Wiggs et al., 2004) by Martin and Kok (2018, MK18, $s \simeq 2210$) are also shown. Moreover, wind tunnel measurements of Θ_t from extrapolating to vanishing transport by Clifton et al. (2006, C06, $s \simeq 830$) are shown. However, since Clifton

et al. (2006) did not feed snow at the tunnel entrance, we have chosen only their data points for freshly fallen snow. Unlike freshly fallen snow, old snow, used for the other measurements by these authors, is very cohesive (Pomeroy & Gray, 1990), and NST of old snow therefore requires a distance to reach equilibrium that is very likely much longer than the length of the wind tunnel of Clifton et al. (2006) in the absence of snow feeding (Comola, Gaume, et al., 2019). However, we have not excluded cohesive measurements if sediment feeding occurred, such as the two windblown sand data points at $Ga \approx 5$ by Bagnold (1937) and Chepil (1945), corresponding to small and thus cohesive mineral grains with $d \approx 75 \mu\text{m}$, and the windblown snow data point by Sugiura et al. (1998), corresponding to potentially very cohesive old snow.

3.1.3. Fluvial Transport Rate Data Sets

Since the model does not capture Q_* for conditions for which $s^{1/2} Ga$ is too small (Figure 5b), we compare it only to subaqueous bedload measurements of Q_* , for which $s^{1/2} Ga$ is sufficiently large. In Figures 5d and 5e, the flume measurements of Q_* by Meyer-Peter and Müller (1948, MPM48, $s \in (2.6, 2.7)$), as corrected by Wong and Parker (2006), for relatively weak driving flows and small slope numbers ($S \approx 0$) and by Smart and Jaeggi (1983, SJ83, $s \in (2.6, 2.7)$) and Capart and Fraccarollo (2011, CF11, $s \approx 1.51$) for relatively intense driving flows and large S are shown. For all these data sets, the applied fluid shear stress is defined as $\tau = \rho g h_m \sin \alpha$ (Appendix A), where h_m is the flow depth, including the sediment-fluid mixture above the bed surface $z = 0$, and we corrected h_m for side wall drag using the method described in Section 2.3 of Guo (2014). For the latter two data sets, the model predictions can depend significantly on α and S for a given Θ . In order to make Q_* only dependent on a single rather than three independent external control parameters, we have approximated Equations 2a and 2b in Figure 5e (but not in Figure 5d). We have used (Appendix F) $\mu_b \approx \mu_b^0 \equiv \lim_{\epsilon_z \rightarrow 1} \mu_b \approx 0.646$, $\Theta_t \approx f_S \Theta_t|_{S=0}$, and $\overline{v_{x*}}_t \approx f_S^{1/2} \overline{v_{x*}}_t|_{S=0}$, where $f_S \equiv 1 - S / \mu_b^0$, and we have used $f_\alpha f_S^{-1} (1 + c_M M_*) \approx 1 + c_M f_\alpha f_S^{-1} M_*$ because the terms $f_\alpha \equiv 1 - \tan \alpha / \mu_b^0$ and f_S predominantly matter when α (and thus M_*) is large. Combined, these approximations yield

$$f_\alpha^2 f_S^{-5/2} Q_* \approx f_\alpha f_S^{-1} M_* \overline{v_{x*}}_t|_{S=0} (1 + c_M f_\alpha f_S^{-1} M_*), \quad (20a)$$

$$f_\alpha f_S^{-1} M_* \approx (f_S^{-1} \Theta - \Theta_t|_{S=0}) / \mu_b^0, \quad (20b)$$

which are expressions independent of α and S for the rescaled transport rate $f_\alpha^2 f_S^{-5/2} Q_*$ as a function of the rescaled Shields number $f_S^{-1} \Theta$. Note that $f_S \approx f_\alpha$ for aeolian NST and slope-driven NST in viscous liquids, implying that this slope correction of Θ and Q_* is consistent with previous results for windblown sand (Iversen & Rasmussen, 1999; Wang et al., 2021), while $f_S < f_\alpha$ for slope-driven NST in turbulent liquids. These differences reflect the different physical origins of f_S and f_α . The physics behind f_S involves the streamwise buoyancy force acting on transported grains, which is associated with the gradient of the viscous fluid shear stress (Maurin et al., 2018), while the physics behind f_α involves the average streamwise fluid momentum balance (Appendix A), which is associated with the gradient of the sum of the viscous and Reynolds-averaged fluid shear stress.

3.1.4. Aeolian Transport Rate Data Sets

For windblown sand and snow, wind tunnel measurements of Q_* by Creyssels et al. (2009, C09, $s \approx 2080$), Ho et al. (2011, H11, $s \approx 2060$), Ralaifarisoa et al. (2020, R20, $s \approx 2210$), and Sugiura et al. (1998, S98, $s \approx 650$) are shown in Figures 5d and 5f. Note that the experiments by Sugiura et al. (1998) were carried out using potentially very cohesive old snow, while the data set by Ralaifarisoa et al. (2020) corresponds to the first controlled measurements of Q_* for intense windblown sand, which are not captured by standard expressions for Q_* from the literature. Furthermore, field measurements of Q_* for windblown sand by Martin and Kok (2017, MK17, $s \approx 2210$) are shown in Figures 5d and 5g. These authors estimated the intermittent (i.e., nonequilibrium) transport rate Q^{in} and the fraction f_Q of active windblown sand, from which we obtained the equilibrium transport rate via $Q = Q^{\text{in}}/f_Q$ (Comola, Kok, et al., 2019).

3.2. NST Regimes

Pähtz and Durán (2018a) provided a criterion to distinguish *bedload*, defined as NST in which a significant portion of transported grains is moving in enduring contacts with the bed surface (e.g., via rolling and sliding), from *saltation*, defined as NST in which this portion is insignificant. This criterion states that saltation

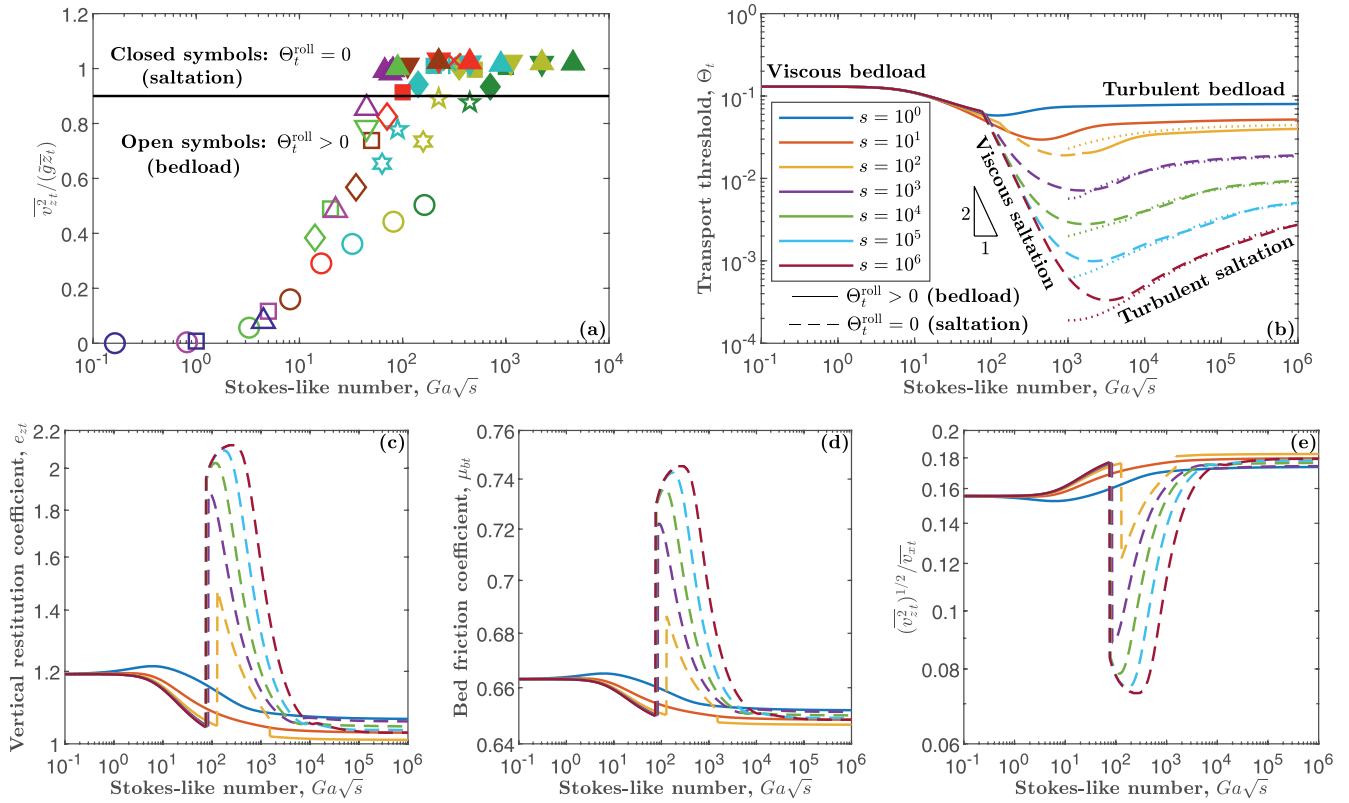


Figure 6. Transport regimes for threshold conditions. (a) Fraction $\bar{v}_{zt}^2 / (\bar{g} \bar{z}_t)$ of transport layer thickness due to the contact-free motion of grains, (b) transport threshold Θ_t , (c) vertical restitution coefficient e_{zt} , (d) bed friction coefficient μ_{bt} , and (e) $(v_{zt}^2)^{1/2} / \bar{v}_{xt}$ versus Stokes-like number $s^{1/2} Ga$. Symbols in (a) correspond to data from DEM-based numerical simulations of NST by Pahltz and Duran (2018a); Pahltz and Duran (2020) for $S = 0$ and varying Ga and s , where the symbol openness indicates the value of Θ_t^{roll} predicted by the transport threshold model using modified parameter values adjusted to the simulations (Section 3.1): $\Theta_t^{\text{roll}} > 0$ (i.e., bedload) for open symbols and $\Theta_t^{\text{roll}} = 0$ (i.e., saltation) for closed symbols. Symbol shapes and colors in (a) are the same as in Figure 5a. The solid line in (a) indicates $\bar{v}_{zt}^2 / (\bar{g} \bar{z}_t) = 0.9$. Solid and dashed lines in (b–e) indicate model predictions for $S = 0$ using the original parameter values. Solid lines indicate a predicted value $\Theta_t^{\text{roll}} > 0$ and dashed lines the predicted value $\Theta_t^{\text{roll}} = 0$. Dotted lines in (b) correspond to the model approximation for turbulent saltation (Equation 24), valid for $s^{1/4} Ga \gtrsim 200$. Figure legends in (c–e) are the same as in (b). DEM, discrete element method; NST, nonsuspended sediment transport.

occurs when more than 90% of the transport layer thickness are due to the contact-free motion of grains: $\bar{v}_{zt}^2 / (\bar{g} \bar{z}_t) \geq 0.9$. Here, for threshold conditions, we distinguish bedload from saltation using the threshold trajectory value of the critical Shields number required for rolling Θ_t^{roll} , which is associated with the rebound energy of the threshold trajectory. When $\Theta_t^{\text{roll}} = 0$, grains are able to escape the pockets of the bed surface solely due to their saltation motion, that is, without the assistance of the near-surface flow. Hence, we identify this regime as saltation and distinguish it from bedload where $\Theta_t^{\text{roll}} > 0$. Figure 6a shows that this criterion is consistent with the one by Pahltz and Duran (2018a) for these authors' data obtained from their DEM-based simulations of NST.

Based on the hop height calculated from the transport threshold model, $H_t = [v_{zt} v_{st} - v_{st}^2 \ln(1 + v_{zt} / v_{st})] / \tilde{g}$ (Equation E3), and the thickness of the viscous sublayer of the turbulent boundary layer, $\delta_{vt} = 10 d / Re_{dt}$, we distinguish between viscous ($H_t \lesssim \delta_{vt}$) and turbulent ($H_t \gtrsim \delta_{vt}$) conditions, giving rise to totally four transport regimes, which are indicated by text in Figure 6b: viscous bedload, turbulent bedload, viscous saltation, and turbulent saltation. It can be seen that viscous bedload occurs when the Stokes-like number $s^{1/2} Ga$ falls below about 75, implying that the validity criterion for the model's transport rate predictions (Equation 19) is only disobeyed for viscous bedload conditions. Note that the transition from viscous bedload to viscous

saltation coincides with a kink in the threshold curves (Figures 5a, 5c, and 6b) and that turbulent saltation occurs when $s^{1/4} Ga \gtrsim 200$ (Section 3.3).

Figure 6b shows that, for viscous saltation, the transport threshold approximately scales as $\Theta_t \propto (s^{1/2} Ga)^{-2}$. To demonstrate the origin of this scaling, we approximate the flow velocity profile in Equation 3 as $u_x \simeq u_* Re_d z/d$, since saltation trajectories are relatively large ($H \gg Z_{\Delta d}$) and fully submerged within the viscous sublayer. Using this profile in Equation 12 and approximating the dimensionless terminal settling velocity by its Stokes drag limit, $v_{s*} \simeq Ga/18$ (Equation 7 for small Ga), yields

$$\text{With vertical drag: } \Theta = 18 \frac{\mu_b(1 + \hat{v}_{\uparrow z}) + \hat{v}_{\uparrow x} - S}{-\hat{v}_{\downarrow z}(1 + \hat{v}_{\uparrow z}) - \hat{v}_{\uparrow z}} (Ga\sqrt{s})^{-2}, \quad (21a)$$

$$\text{Neglected vertical drag: } \Theta \simeq 18 \frac{\mu_b^o(1 + \hat{v}_{\uparrow z}) + \hat{v}_{\uparrow x} - S}{\frac{1}{3}\hat{v}_{\uparrow z}^2} (Ga\sqrt{s})^{-2}, \quad (21b)$$

where Equation 21b is the approximation of Equation 21a valid for negligible vertical drag (Appendix F). After linking $\hat{v}_{\downarrow z}$, $\hat{v}_{\uparrow x}$, $\hat{v}_{\downarrow x}$, and μ_b to $\hat{v}_{\uparrow z}$ via Equations 11, 13a, and 13c, the crucial difference between both equations is that Θ in Equation 21a first monotonously decreases with $\hat{v}_{\uparrow z}$ until it approaches a minimum and then monotonously increases with $\hat{v}_{\uparrow z}$, whereas Θ in Equation 21b monotonously decreases with $\hat{v}_{\uparrow z}$ for the entire range of $\hat{v}_{\uparrow z}$. Hence, obtaining the transport threshold Θ_t from Equation 21a via minimizing Θ (Equation 16) yields $\Theta_t \propto (s^{1/2} Ga)^{-2}$ for a fixed S , whereas the use of Equation 21b would yield a contradiction: an infinitely large threshold trajectory ($\hat{v}_{\uparrow z} = \infty$) and $\Theta_t = 0$. Hence, vertical drag is not negligible in viscous saltation. In fact, the spikes in Figure 6c indicate a substantial positive deviation from $e_z = 1$ for viscous saltation, whereas e_z is close to unity for the other regimes. Larger values of e_z mean that vertical drag suppresses the vertical motion of grains more strongly, causing a slight positive deviation from $\mu_{bt} = \text{const}$ (spikes in Figure 6d) and a substantial negative deviation from $(v_{zt}^2)^{1/2} \propto \bar{v}_{xt}$ (spikes in Figure 6e). In Appendix C, we use these two scaling relations and their violation for viscous saltation to justify the rebound boundary conditions.

We now perform a similar analysis for turbulent saltation. We approximate the flow velocity profile in Equation 3 as the log-layer profile $u_x \simeq \kappa^{-1} u_* \ln(z/z_o)$, where z_o is the roughness height, since saltation trajectories are relatively large ($H \gg Z_{\Delta d}$) and substantially exceed the viscous sublayer and buffer layer. Using this profile in Equation 12 and neglecting vertical drag (as $e_z \simeq 1$, see Figure 6c) yields

$$\frac{\sqrt{\Theta}}{\kappa v_{s*}^o} \ln \left(\frac{s v_{s*}^{o2} \hat{v}_{\uparrow z}^2}{3 z_o / d} \right) \simeq \mu_b^o(1 + \hat{v}_{\uparrow z}) + \hat{v}_{\uparrow x} - S, \quad (22)$$

where $v_{s*}^o \equiv \lim_{e_z \rightarrow 1} v_{s*} = v_{s*}^o |_{\mu_b = \mu_b^o}$. Using that $\hat{v}_{\uparrow x} \propto \hat{v}_{\uparrow z}$ for conditions in which vertical drag is negligible (Appendix F), we can minimize Θ in Equation 22 via $d\Theta / d\hat{v}_{\uparrow z} = 0$ to obtain Θ_t (Equation 16). This yields after some rearrangement $\mu_b^o \hat{v}_{\uparrow z} + \hat{v}_{\uparrow x} = 2\sqrt{\Theta_t} / (\kappa v_{s*}^o)$ and subsequently, using the definition of the hat,

$$\frac{\sqrt{\Theta_t}}{\kappa} \left[\ln \left(\frac{\beta s \Theta_t}{\kappa^2 z_{ot} / d} \right) - 2 \right] \simeq (\mu_b^o - S) v_{s*}^o, \quad (23)$$

where $\beta \equiv \lim_{e_z \rightarrow 1} (4 v_{\uparrow z}^2 / 3) / (\mu_b v_{\uparrow z} + v_{\uparrow x})^2 \simeq 0.136$ (Appendix G). For both $z_{ot} = d/30$, valid in the fully rough regime ($Re_d \gtrsim 70$), and $z_{ot} = d/(9 Re_{dt})$, valid in the fully smooth regime ($Re_d \lesssim 4$), Equation 23 can be explicitly solved (Appendix G). The maximum of both solutions (dotted lines in Figure 6b) well approximates Θ_t for turbulent saltation:

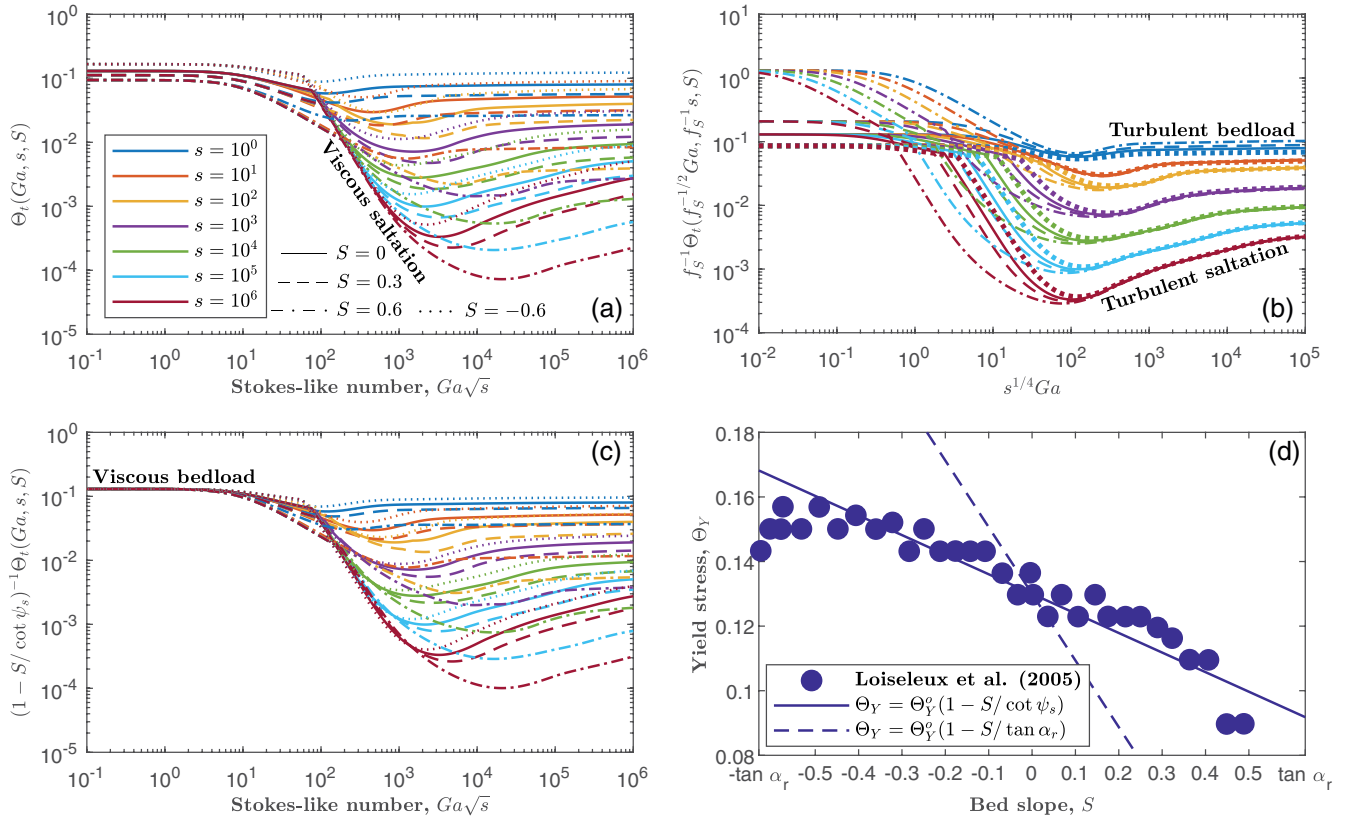


Figure 7. Effect of slope number S on transport threshold predictions. (a) Transport threshold $\Theta_t(Ga, s, S)$ versus Stokes-like number $s^{1/2} Ga$. (b) Rescaled transport threshold $f_s^{-1}\Theta_t(f_s^{-1/2}Ga, f_s^{-1}s, S)$, where $f_s = 1 - \mu_b^o / S$, versus the dimensionless number $s^{1/4} Ga$, which roughly controls the occurrence of the transport threshold minimum for saltation (Pähtz & Durán, 2018a). Note that, according to Equation 25b, $f_s^{-1}\Theta_t(f_s^{-1/2}Ga, f_s^{-1}s, S)$ is approximately equal to $\Theta_t(Ga, s, 0)$ (independent of S) for turbulent NST. (c) Rescaled transport threshold $(1 - S / \cot \psi_s)^{-1}\Theta_t(Ga, s, S)$ versus $s^{1/2} Ga$. (d) Yield stress Θ_Y versus S . Lines in (a–c) correspond to model predictions. Symbols in (d) correspond to measurements of Θ_t by Loiseleux et al. (2005) for $Ga = [0.17, 0.54]$ and $s = 2.5$ (i.e., $\Theta_t \simeq \Theta_Y$) and various S . Measurements of the angle of repose α_r by Loiseleux et al. (2005) are indicated in the abscissa of (d). The solid line in (d) corresponds to the slope correction predicted by the model for viscous bedload (Equation 25c). The dashed line in (d) corresponds to the classical slope correction. Figure legends in (b) and (c) are the same as in (a). NST, nonsuspended sediment transport.

$$\begin{aligned} \Theta_t &\simeq \max(\Theta_t^{\text{rough}}, \Theta_t^{\text{smooth}}), \text{ with} \\ \Theta_t^{\text{rough}} &\equiv \left(\frac{\kappa(\mu_b^o - S)v_{s*}^o}{2W[(\mu_b^o - S)v_{s*}^o \sqrt{15e^{-2}\beta s / 2}]} \right)^2 \text{ and} \\ \Theta_t^{\text{smooth}} &\equiv \left(\frac{\kappa(\mu_b^o - S)v_{s*}^o}{3W[(\mu_b^o - S)v_{s*}^o \sqrt[3]{e^{-2}\kappa\beta s Ga / 3}]} \right)^2. \end{aligned} \quad (24)$$

3.3. Bed Slope Dependency of Transport Threshold

Figure 7 shows how the slope number S affects the transport threshold predictions.

For the different NST regimes, different scaling laws are found:

$$\text{Viscous saltation: } \Theta_t(Ga, s, S) \simeq \Theta_t(Ga, s, 0), \quad (25a)$$

$$\text{Turbulent NST: } \Theta_t(Ga, s, S) = f_S \Theta_t(\sqrt{f_S} Ga, f_S s, 0), \quad \text{and} \quad (25b)$$

$$\text{Viscous bedload: } \Theta_t(Ga, s, S) = \Theta_Y = (1 - S / \cot \psi_s) \Theta_Y^0, \quad (25c)$$

the validity of which is shown in Figures 7a–7c, respectively. Equation 25a follows from the fact that the term $\mu_b(1 + \hat{v}_{\uparrow z}) + \hat{v}_{\uparrow z}$ in Equation 21a is substantially larger than S for the threshold trajectory, since grain velocities become comparable to the terminal settling velocity v_s (i.e., $\hat{v}_{\uparrow z} \sim 1$) because of vertical drag in viscous saltation, implying that the effect of S is small. In contrast, for turbulent NST, vertical drag can be neglected (i.e., $\hat{v}_{\uparrow z} \ll 1$), which leads to Equation 25b (Appendix F). Note that, for turbulent saltation, the validity of Equation 25b requires $s^{1/4} Ga \gtrsim 200$, where $s^{1/4} Ga$ is a dimensionless number that roughly controls the occurrence of the transport threshold minimum for saltation (Pähtz & Durán, 2018a), consistent with Figure 7b. Equation 25c follows from Equation 15 as, for viscous bedload, $\frac{1}{2} s v_s^2 \hat{v}_{\uparrow}^2 \propto Ga^2 s$ becomes negligible (Section 2.3.3).

Classically, the term $1 - S/\tan \alpha_r$, where α_r is the angle of repose (i.e., the slope angle at which the entire granular bed, as opposed to only the bed surface, fails), is used to correct Θ and/or Θ_t in NST (Iversen & Rasmussen, 1994; Maurin et al., 2018), in reasonable agreement with threshold measurements (Chiew & Parker, 1994; Iversen & Rasmussen, 1994). Both the slope correction term $f_S \equiv 1 - S / \mu_b^0$ in Equation 25b and the slope correction term $1 - S/\cot \psi_s$ in Equation 25c resemble the functional structure of $1 - S/\tan \alpha_r$. However, μ_b^0 is a purely kinematic quantity and entirely unrelated to $\tan \alpha_r$ even though its value ($\mu_b^0 \approx 0.646$) is very close to typical values of $\tan \alpha_r$. Likewise, $\cot \psi_s \approx 2.14$ is substantially larger than typical values of $\tan \alpha_r$ and the predicted slope correction for viscous bedload therefore much milder than the classical one. Consistently, Figure 7d shows that, for the viscous bedload experiments by Loiseleux et al. (2005), the model (solid line) reproduces the measured behavior that Θ_t changes only mildly with S for $|S| \lesssim 0.5$, while the classical slope correction (dashed line) fails to capture these data. The deviations between model and measurements for $|S| \gtrsim 0.5$ are likely due to the fact that $|S|$ approaches $\tan \alpha_r$, weakening the resistance of the bulk of the bed. In fact, once the bulk of the bed is close to yield, this will probably affect the resistance of bed surface grains via long-range correlations (since yielding is probably a critical phenomenon, Pähtz et al., 2020), which the model does not account for.

Lastly, we emphasize that the model predictions do not take into account that large bed slopes in nature (e.g., for mountain streams) are usually accompanied by very small flow depths of the order of $1d$, which cause bed mobility to decrease rather than increase with S (Prancevic et al., 2014; Prancevic & Lamb, 2015).

4. Discussion

4.1. Transport Threshold Interpretation

In Section 2, we first idealized equilibrium NST in a manner that eliminates all kinds of grain trajectory fluctuations. We then defined the transport threshold Θ_t as the smallest Shields number for which a nontrivial steady state grain trajectory exists. This definition raises two important questions:

- What values of the initial lift-off velocity $\mathbf{v}_{\uparrow}^0$ are sufficient for a grain to approach the nontrivial steady state threshold trajectory?
- How sensitive is this threshold trajectory to trajectory fluctuations?

This section gives answers to these questions and discusses consequences for the physical interpretation of Θ_t arising from these answers.

We reiterate that, for most bedload conditions, the model predicts $\Theta_t = \Theta_t^{\text{roll}}$ (Section 2.4). This implies that the average flow is barely able to sustain a rolling motion of grains out of the most stable pockets of the bed surface, meaning that grain motion would stop for an arbitrarily small negative fluctuation of the threshold rebound velocity $\mathbf{v}_{\text{rt}}$, since smaller grain energies make rolling out of the most stable bed surface pockets more difficult (Section 2.3.3).

For saltation, the model predicts a similar behavior as for bedload. To show this, we calculate the saltation trajectory evolution from equations E1a, E1b, E1d, and E1f using Equation 3 and the periodic saltation trajectory (i.e., steady) value of the dimensionless terminal settling velocity v_{s*} , the calculation of which was described in Section 2.4. Based on this calculation, Figure 8a shows for an exemplary saltation case ($s = 2000$, $Ga = 40$, $S = 0$, and $\Theta \simeq [1.0, 1.2, 2.0]\Theta_t$) the critical lines $\hat{v}_{\uparrow c}^0$ separating those conditions with an initial dimensionless lift-off velocity $\hat{v}_{\uparrow}^0$ that approach the periodic trajectory solution $\hat{v}_{\uparrow}$ (supercritical, northeast of the critical lines, see solid lines in Figures 8c and 8d for an exemplary trajectory) from those conditions that approach no motion (subcritical, southwest of the critical lines, see dashed line in Figure 8c for an exemplary trajectory). Furthermore, Figure 8b shows that the distance $\min_{\hat{v}_{\uparrow c}^0} |\hat{v}_{\uparrow} - \hat{v}_{\uparrow c}^0|$ between $\hat{v}_{\uparrow}$ and $\hat{v}_{\uparrow c}^0$ obeys critical scaling behavior for sufficiently small $\Theta - \Theta_t$ and vanishes in the limit $\Theta \rightarrow \Theta_t$. Hence, arbitrarily small negative fluctuations of $\mathbf{v}_{\uparrow t}$ will cause saltation to cease in time.

The model prediction that both bedload and saltation threshold trajectories are unstable against arbitrarily small negative fluctuations of $\mathbf{v}_{\uparrow t}$ implies that, for realistic natural settings associated with substantial grain trajectory fluctuations, continuous NST cannot be sustained once Θ is too close to Θ_t . In other words, Θ_t must be strictly smaller than the continuous transport threshold Θ^{cont} regardless of how continuous NST is defined (an unambiguous definition is currently an open problem, Pähtz et al., 2020, Section 4.1.3.2).

This subsection is subdivided into further subsections. Section 4.1.1 models Θ^{cont} for aeolian NST and compares the model predictions with experimental data. Section 4.1.2 discusses intermittent NST dynamics occurring for $\Theta < \Theta^{\text{cont}}$.

4.1.1. Continuous Transport Threshold

Understanding the behavior of Θ^{cont} is crucial because equilibrium transport rate expressions, such as Equations 2a and 2b, are invalid for intermittent (i.e., nonequilibrium) NST conditions (Comola, Kok, et al., 2019; Pähtz et al., 2020). To this end, let us consider NST conditions with $\Theta > \Theta_t$ and suppose that the system departs more and more from the equilibrium by depositing grains on the bed surface. The more grains are deposited, the more the flow will be undisturbed by the presence of transported grains. To drive this system back to equilibrium, it is required that bed surface grains are entrained and subsequently net accelerated by the undisturbed flow. We therefore propose that Θ^{cont} corresponds to the minimal Shields number for which the average velocity of such an entrained grain $\mathbf{v}_{\uparrow}^e$ is supercritical, denoted by $\mathbf{v}_{\uparrow}^e \succeq \mathbf{v}_{\uparrow c}^0$ (note that both $\mathbf{v}_{\uparrow}^e$ and $\mathbf{v}_{\uparrow c}^0$ depend on the undisturbed flow condition (Ga, s, S, Θ)):

$$\Theta^{\text{cont}}(Ga, s, S) \equiv \min_{\hat{v}_{\uparrow z} : \mathbf{v}_{\uparrow}^e(Ga, s, S, \Theta) \succeq \mathbf{v}_{\uparrow c}^0(Ga, s, S, \Theta)} \Theta(Ga, s, S, \hat{v}_{\uparrow z}), \quad (26)$$

where $\Theta(Ga, s, S, \hat{v}_{\uparrow z})$ denotes, like before (Section 2.4), a periodic trajectory solution. Consistent with this definition, Figure 8a shows that, for saltation, the range of supercritical initial dimensionless lift-off velocities ($\hat{v}_{\uparrow}^0 \succeq \hat{v}_{\uparrow c}^0$) substantially increases with Θ/Θ_t . Note that our proposed definition of Θ^{cont} is similar to the one by Doorschot and Lehning (2002). The main and possibly only difference is that these authors' definition referred to the average grain lifting off from the bed surface, including entrained and rebounding grains, rather than only the average entrained grain.

For aeolian NST, we can model Θ^{cont} using Equation 26 and the fact that, on average, bed surface grains entrained by impacts of transported grains are more energetic than grains entrained directly by the flow and therefore more likely to be supercritical. In fact, the former grains are literally ejected and their average ejection velocity $\mathbf{v}_{\uparrow}^e$ is weakly but significantly correlated with the impact velocity $\mathbf{v}_i$ (Beladjine et al., 2007). An empirical scaling relation that is consistent with experimental data is

$$v_{\uparrow x}^e = 0, \quad (27a)$$

$$v_{\uparrow z}^e / \sqrt{gd} = c_e (|\mathbf{v}_i| / \sqrt{gd})^{1/4}, \quad (27b)$$

where c_e is a proportionality constant close to unity (Beladjine et al., 2007). Figure 8e shows model predictions of Θ^{cont} , using Equations 26, 27a, and 27b with $c_e = 0.8$, and compares them with measurements of

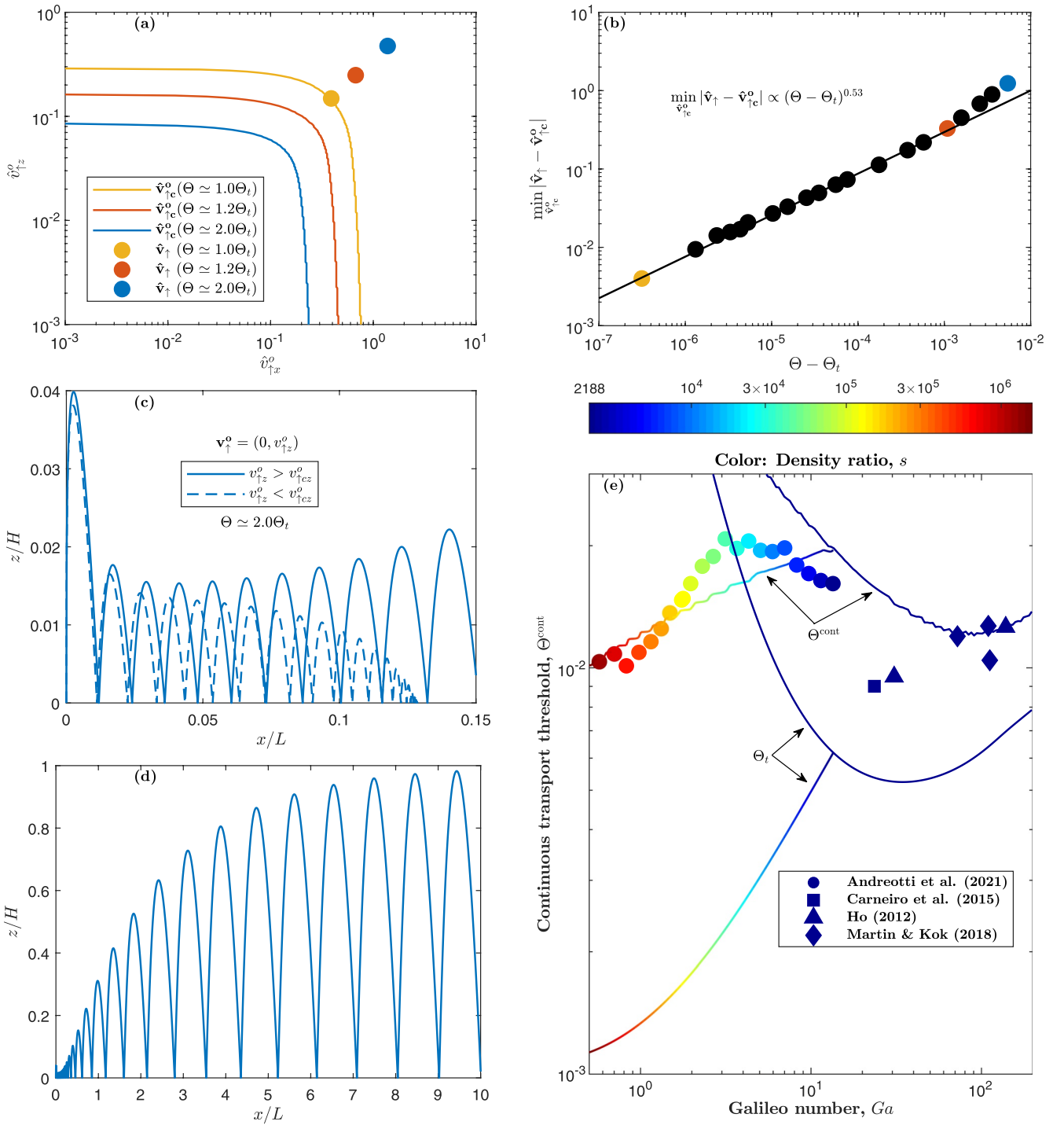


Figure 8. Grain trajectories and continuous transport threshold Θ^{cont} . (a) Critical lines $\hat{v}_{\uparrow c}^o$ separating supercritical conditions (northeast of the lines) from subcritical conditions (southwest of the lines) exemplary for the case $s = 2000$, $Ga = 40$, $S = 0$, and three different Shields numbers Θ . Supercritical refers to initial conditions ($\hat{v}_{\uparrow}^o$) that approach the periodic saltation trajectory solution $\hat{v}_{\uparrow}$ (circles). Subcritical refers to initial conditions that approach no motion. (b) Critical scaling of the distance $\min_{\hat{v}_{\uparrow c}^o} |\hat{v}_{\uparrow} - \hat{v}_{\uparrow c}^o|$ between $\hat{v}_{\uparrow}$ and $\hat{v}_{\uparrow c}^o$. (c) Examples for grain trajectories $z(x)$, where z and x are normalized by the hop height H and hop length L , respectively, of the periodic saltation trajectory, with supercritical (solid lines) and subcritical (dashed lines) initial conditions $\mathbf{v}_{\uparrow}^o = (0, v_{\uparrow z}^o)$ for $\Theta \approx 2.0\Theta_t$. (d) The same plot as in (c) but for a much wider range of x/L . (e) Θ^{cont} versus Galileo number Ga . Symbols in (e) correspond to windblown sand measurements of Θ^{cont} for either $d = 125 \mu\text{m}$ and varying ρ_f (Andreotti et al., 2021) or $\rho_f = 1.2 \text{ kg/m}^3$ and varying d (Carneiro et al., 2015; Ho, 2012; Martin & Kok, 2018). The symbol color indicates the value of the density ratio s . Lines in (e) correspond to model predictions of Θ^{cont} and the transport threshold Θ_t .

Θ^{cont} for aeolian NST (Andreotti et al., 2021; Carneiro et al., 2015; Ho, 2012; Martin & Kok, 2018). Note that Andreotti et al. (2021) carried out their experiments in a wind tunnel with varying air pressure and defined threshold conditions as the transition between saltation of groups of particles (bursts) to intermittent saltation of single particles (at high pressure) or no transport (at low pressure). These authors interpreted their measurements as measurements of the transport threshold Θ_t , that is, they assumed $\Theta^{\text{cont}} = \Theta_t$. However, Figure 8e shows that, for many of their measured conditions, the model predictions of Θ_t are nearly an order of magnitude below those of Θ^{cont} , and only the latter are consistent with the measurements.

4.1.2. Intermittent NST

For Shields numbers Θ below Θ^{cont} , NST may remain intermittent. There are two distinct kinds of transport intermittency. The first kind occurs when turbulence-driven bed sediment entrainment events associated with energetic turbulent eddies (Cameron et al., 2020; Paterna et al., 2016; Valyrakis et al., 2011; Zheng et al., 2020) generate intermittent rolling events of entrained grains (Pähtz et al., 2020). This kind of intermittency is negligible for saltation, since the transport rate of rolling grains is much smaller than that of saltating grains. However, it is important for bedload, where it is known to occur also below Θ_t (Pähtz et al., 2020). However, since the average flow cannot sustain the average motion of grains, such rolling events end very quickly below Θ_t . Second, turbulent fluctuation events that temporarily push Θ^{fluc} above Θ^{cont} cause a different kind of intermittency, which is usually associated with saltation (Comola, Kok, et al., 2019), though the exact mechanism of this intermittency depends on the physical processes behind Θ^{cont} (Pähtz et al., 2020). In the context of our proposed definition of Θ^{cont} , such events will cause grains entrained by grain-bed impacts to approach a periodic saltation trajectory, thus generating a saltation chain reaction that rapidly increases the transport rate Q . Once the turbulent fluctuation event is over, provided that $\Theta > \Theta_t$, saltation may maintain a large value of Q for a relatively long time (Pähtz et al., 2020), though not indefinitely as trajectory fluctuations will cause grains to eventually settle (Section 4.1). Nonetheless, this leads to a substantial hysteresis of Q for $\Theta_t < \Theta < \Theta^{\text{cont}}$ (Carneiro et al., 2015).

Lastly, we emphasize that, although turbulence plays a crucial role for the complex intermittent behavior of NST for $\Theta_t < \Theta < \Theta^{\text{cont}}$, both Θ_t and Θ^{cont} are statistical quantities referring to the grain motion averaged over long times. That is, the influence of turbulence on Θ_t and Θ^{cont} is probably relatively weak.

4.2. Importance of Cohesion

Despite not explicitly accounting for cohesion, the coupled model captures measurements of the transport threshold Θ_t for small ($d \approx 75 \mu\text{m}$), that is, cohesive windblown sand grains by Bagnold (1937) and Chepil (1945) and measurements of Θ_t and the dimensionless transport rate Q_* for potentially very cohesive windblown snow by Sugiura et al. (1998), as shown in Figures 5c, 5d, and 5f. In particular, the model suggests that the increase of Θ_t with decreasing grain size d for sufficiently small d , which was previously attributed to cohesion (e.g., Berzi et al., 2017; Shao & Lu, 2000), is solely due to NST entering the viscous saltation regime. In this regime, $\Theta_t \propto (s^{1/2}Ga)^{-2} \sim d^{-3}$, which is a stronger decrease than the one ($\Theta_t \sim d^{-2}$) predicted by standard cohesion-based models (Berzi et al., 2017; Shao & Lu, 2000).

The agreement of the model with cohesive data supports the controversial suggestion by Comola, Gaume, et al. (2019) that, for equilibrium aeolian NST, Θ_t and Q_* are nearly unaffected by the strength of cohesive bonds between bed grains. Further support comes from the model conceptualization. In fact, for the saltation regime, to which aeolian NST belongs, the only manner in which bed grains affect the model conceptualization is via the rebound laws, since both Q_* (Pähtz & Durán, 2020) and Θ_t (Section 2) have been conceptually introduced as bed sediment entrainment-independent physical quantities. However, the rebound laws are insensitive to the strength of cohesive bonds between bed grains for the simulation data by Comola, Gaume, et al. (2019) (Figures 2d and 2e). Note that the DEM-based numerical simulations of equilibrium aeolian NST by Comola, Gaume, et al. (2019) were limited to Earth's atmospheric conditions, for which saltating grains perform relatively large hops on average ($10 \dots 100d$, see Ho et al., 2014, Table A4). However, our model conceptualization suggests that the insensitivity of equilibrium NST to cohesion is not limited to Earth's atmospheric conditions but applies to equilibrium NST in general as long as transported grains are unable to form cohesive bonds with bed grains during their contacts with the bed. This requirement is probably satisfied in the saltation regime (dashed lines in Figure 6b) as the duration of grain-bed rebounds

is very short. However, it may be violated in the bedload regime (solid lines in Figure 6b) as grains tend to move in enduring contact with the bed. We therefore propose that the effects of cohesion tend to become negligible once the model predicts $\Theta_t^{\text{roll}} = 0$, which is the criterion with which we identify the saltation regime (Figure 6a). However, we recognize that this proposition is controversial.

5. Conclusions

In this study, we have proposed and evaluated a model of the two arguably most important statistical properties of equilibrium NST: the transport threshold Shields number Θ_t and dimensionless transport rate Q^* . The model captures, within a factor of about 2, experimental and numerical data of Θ_t across the entire range of environmental conditions and experimental and numerical data of Q^* across weak and intense conditions in which a small critical value of the Stokes-like number $s^{1/2} Ga$ (Equation 19) is exceeded (Figure 5). Such conditions include *subaqueous bedload*, *windblown sand*, and *windblown snow*. The conditions that are not captured by the transport rate model correspond solely to *viscous bedload* (Figure 6b). Note that the agreement between model and experimental data includes the first controlled measurements of Q^* for intense windblown sand by Ralaiaisoa et al. (2020), which are not captured by standard expressions for Q^* from the literature.

The agreement between the model and experimental data is not the result of parameter adjustment. In fact, all model parameters have been obtained from independent data sets: c_M and Z_Δ from DEM-based numerical simulations of NST (Pähtz & Durán, 2018a, 2020), A , B , and C from grain-bed rebound experiments (Beladjine et al., 2007), and Θ_y^o and ψ_s from experiments and numerical simulations of bed yielding and grain entrainment driven by laminar flows (Agudo et al., 2017; Charru et al., 2004; Houssais et al., 2015; Loiseleux et al., 2005; Ouriemi et al., 2007).

NST is a highly fluctuating physical process (Ancey, 2020b; Durán et al., 2011) because of turbulence, surface inhomogeneities, and variations of grain size and shape and packing geometry. Furthermore, the energy of transported grains varies strongly due to variations of their flow exposure duration since their entrainment from the bed (Durán et al., 2011). However, such internal variability is completely neglected in the model, since it represents equilibrium NST by grains moving in identical periodic saltation trajectories. The high predictive capability of the model therefore suggests that crucial statistical properties of NST are relatively insensitive to its internal variability.

Although the model represents threshold conditions by a continuous grain motion, we have shown that Θ_t must be strictly smaller than the continuous transport threshold Θ^{cont} for realistic natural settings (Section 4.1). In particular, a semiempirical extension of the model for aeolian NST predicts values of Θ^{cont} that are consistent with measurements (Figure 8e). For $\Theta < \Theta^{\text{cont}}$, NST can exhibit complex intermittency characteristics.

The model straightforwardly provides a criterion, which we validated with numerical data from the literature (Figure 6a), that distinguishes *bedload*, defined as NST in which a significant portion of transported grains is moving in enduring contacts with the bed surface (e.g., via rolling and sliding), from *saltation*, defined as NST in which this portion is insignificant (Section 3.2). Based on the conceptualization of the model, we have proposed that, in the saltation regime, equilibrium NST is insensitive to the cohesiveness of bed grains (Section 4.2). Consistently, the cohesionless model captures measurements for aeolian saltation of cohesive sand and snow grains. In particular, the increase of Θ_t with decreasing grain size d for wind-blown sand with $d \lesssim 100 \mu\text{m}$, previously attributed to cohesion (e.g., Berzi et al., 2017; Shao & Lu, 2000), is predicted to be solely caused by NST entering the *viscous saltation* regime (Figure 6b), corresponding to saltation within the viscous sublayer of the turbulent boundary layer. In this regime, the model predicts $\Theta_t \propto (s^{1/2} Ga)^{-2} \sim d^{-3}$, which is a stronger decrease than the one ($\Theta_t \sim d^{-2}$) predicted by standard cohesion-based models (Berzi et al., 2017; Shao & Lu, 2000). Hence, the model supports the controversial notion that the strength of cohesive bonds between bed grains does neither significantly affect Θ_t nor Q^* , which Comola, Gaume, et al. (2019) suggested based on their DEM-based numerical simulations of equilibrium aeolian NST for Earth's atmospheric conditions.

Classically, the transport threshold has been corrected for a nonzero slope number S via $\Theta_t = (1 - S/\tan \alpha_r)$ $\Theta_t|_{S=0}$, where α_r is the angle of repose (Iversen & Rasmussen, 1994; Maurin et al., 2018). However, the model predicts that the predominant slope correction factor for turbulent NST is actually $1 - S / \mu_b^o$ (Equation 25b). Although $\mu_b^o = 0.646$ is very close to typical values of $\tan \alpha_r$, its physical meaning in the model is fundamentally different. It is a purely kinematic bed friction coefficient associated with the laws that describe a grain-bed rebound. Furthermore, for viscous bedload, the model predicts a much milder bed slope dependency than the classical one (Equation 25c), in agreement with measurements (Figure 7d). Note that the model predictions do not take into account that large bed slopes in nature (e.g., for mountain streams) are usually accompanied by very small flow depths of the order of $1d$, which cause bed mobility to decrease rather than increase with S (Prancevic et al., 2014; Prancevic & Lamb, 2015).

Both the insensitivity of Θ_t to cohesion and the insensitivity of its slope dependency to α_r are features of the fact that Θ_t in the model is *not* associated with the entrainment of bed surface grains, neither by the flow nor grain-bed impacts. Instead, Θ_t in the model can be characterized as a *rebound threshold*, as defined and extensively discussed in a recent review on the matter (Pähtz et al., 2020).

In the future, the model may be used to reliably predict equilibrium NST in extraterrestrial environments, such as on Venus, Titan, Mars, and Pluto. However, while the model can probably be applied to Venus and Titan conditions, since they are well within the range of environmental conditions for which we evaluated the model, the application of the model to conditions with very large density ratios $s \gtrsim 10^4$ (e.g., Mars and Pluto) requires further model validation. For this reason, DEM-based numerical simulations of NST for conditions with large s are planned in the future.

Appendix A: Derivation of Equation 2b

Pähtz and Durán (2020) derived their equation S20, which is the analog to Equation 2b, from the fluid and particle momentum balances for equilibrium conditions:

$$\frac{d\tau_f}{dz} = f_x, \quad \text{with} \quad \tau_f(z \geq z_{\max}) = \tau, \quad (\text{A1a})$$

$$\frac{d\tau_g}{dz} = -f_x - \rho g \sin \alpha, \quad (\text{A1b})$$

$$\frac{dp_g}{dz} = -\rho \tilde{g}, \quad (\text{A1c})$$

where τ_f is the fluid shear stress disturbed by the grain motion, τ_g the particle shear stress, p_g the vertical particle pressure, f_x the streamwise fluid-particle interaction force per unit volume (consisting of fluid drag and buoyancy in this paper), ρ the particle concentration, and $z_{\max}$ the top of the transport layer, which no grain exceeds (i.e., $f_x = \rho = 0$ for $z \geq z_{\max}$). Here, we argue for an improvement of Equation A1a, which is the streamwise fluid momentum balance using the inner turbulent boundary layer approximation.

Let us consider the fluid momentum balance, without boundary layer approximation, for slope-driven equilibrium NST. It reads (Maurin et al., 2018)

$$\frac{d\tau_f}{dz} = -(\rho_f - s^{-1}\rho)g \sin \alpha + f_x, \quad \text{with} \quad \tau_f(z = h_m) = 0, \quad (\text{A2})$$

where h_m is the elevation of the free surface relative to the bed surface elevation $z = 0$, defined in Figure 1 (i.e., h_m is the flow depth, including the sediment-fluid mixture above the bed surface). Integrating Equation A2 from $z = 0$ to $z = h_m$ yields

$$\tau - \tau_f(0) \equiv \rho_f g h_m \sin \alpha - \tau_f(0) = \int_0^{z_{\max}} (f_x + s^{-1}\rho g \sin \alpha) dz. \quad (\text{A3})$$

Hence, for the inner turbulent boundary layer approximation to be consistent with such slope-driven conditions, Equation A1a should be modified to

$$\frac{d\tau_f}{dz} = f_x + s^{-1}\rho g \sin \alpha, \quad \text{with} \quad \tau_f(z \geq z_{\max}) = \tau. \quad (\text{A4})$$

Integrating Equation A1b, A1c, and A4 from $z = 0$ to $z = z_{\max}$, assuming $\tau - \tau_f(0) \simeq \tau - \tau_t$ (Pähtz & Durán, 2020) and using $M \equiv \int_0^{z_{\max}} \rho dz$, then yields

$$\tau - \tau_t \simeq \tau - \tau_f(0) = \int_0^{z_{\max}} f_x dz + s^{-1}Mg \sin \alpha, \quad (\text{A5a})$$

$$\tau_g(0) = \int_0^{z_{\max}} f_x dz + Mg \sin \alpha, \quad (\text{A5b})$$

$$p_g(0) \simeq M\tilde{g}. \quad (\text{A5c})$$

Combining these equations, using $\mu_b \equiv \tau_g(0)/p_g(0)$, yields Equation 2b.

Appendix B: The Mean Flow Velocity Assumption

We assume that the mean motion of grains driven by a fluctuating turbulent flow is the same as the mean motion of grains driven by a mean turbulent flow (Section 2.1.1). To be approximately satisfied, this assumption has two requirements. First, turbulent diffusion must be negligible, otherwise, turbulent ejection events exercise a substantial control on the mean motion of grains (Aksamit & Pomeroy, 2018; Lelouvetel et al., 2009). This requirement is satisfied because we consider only fully NST (i.e., Rouse number $Ro \gtrsim 2.8$, Naqshband et al., 2017). Second, the ratio between the standard deviation σ_τ and mean τ of the fluctuating applied fluid shear stress τ^{fluc} should be sufficiently small so that streamwise grain velocity fluctuations are dominated by the randomness caused by interactions with the bed surface rather than fluid shear stress fluctuations. Note that we are modeling only those grains that have approached a nontrivial steady state trajectory, that is, comparably energetic grains that have survived multiple interactions with the bed surface without being captured. The velocity distribution of such grains would be expected to be Gaussian if their velocity fluctuations were predominantly caused by grain-bed interactions (Ho et al., 2012), while it would be expected to be skewed (e.g., log-normal or exponential) if their velocity fluctuations were predominantly caused by streamwise flow fluctuations (Shim & Duan, 2019) because τ^{fluc} is log-normally distributed (Cheng & Law, 2003; Martin et al., 2013). Coupled DEM/large eddy simulations by Liu et al. (2019) of a grain saltating along a fixed quadratically arranged (i.e., idealized) bed driven by water showed a skewed streamwise velocity distribution and, consistently, a substantial difference in the average streamwise grain motion when compared with simulations in which turbulent fluctuations were turned off. In contrast, experiments indicate symmetric Gaussian-like distributions of the streamwise velocity of energetic grains in natural aeolian and fluvial NST along random close packed (i.e., nonidealized) erodible beds (Ancy & Pascal, 2020; Heyman et al., 2016; Kang et al., 2008; Shim & Duan, 2019). We take this as evidence that, for natural conditions, the second requirement is approximately satisfied, though we recognize the potential of making a substantial error when assuming that the fluctuating turbulent flow can be approximated by its mean for modeling the mean motion of energetic grains.

Appendix C: From Bed Friction Law to Periodic Saltation With Rebound Boundary Conditions

This appendix presents justifications, partially based on the bed friction law in Section 2.2, for why one can represent the entire grain motion in NST, including grains that roll and/or slide along the bed, by a periodic saltation motion with rebound boundary conditions. First, we justify representing NST by a pure saltation motion (section C1). Second, we justify the use of rebound boundary conditions (section C2).

Appendix C1: Justification for Representing NST by a Pure Saltation Motion

Like previous studies (e.g., Bagnold, 1973; Charru et al., 2004), we assume that one can represent the entire grain motion in NST, including grains that roll and/or slide along the bed, by a pure saltation motion.

A heuristic justification for this assumption is that bed grains that initially roll after their entrainment from the bed surface very quickly begin to make very small hops due to the geometrical disorder of the bed (e.g., see Heyman et al., 2016, Movie S3). A more physical justification of this assumption was provided by Pähz and Durán (2018b) based on the bed friction law in Section 2.2. To explain these authors' findings, we introduce their precise definition of the bed friction coefficient: $\mu_b \equiv \tau_g(0)/p_g(0)$, where τ_g is the particle shear stress and p_g the vertical particle pressure. Both $\tau_g \equiv \tau_g^c - \rho \langle v'_z v'_x \rangle$ and $p_g \equiv p_g^c + \rho \langle v'^2_z \rangle$ are separated into a contribution from particle contacts (superscript c) and a kinetic contribution associated with the grain fluctuation motion between contacts ($\pm \rho \langle v'_z v'_i \rangle$). Pähz and Durán (2018b) found that, for a wide range of DEM-based numerical simulations of NST, the kinetic contributions dominate when most grains move in saltation (typical for aeolian NST), while the contact contributions dominate when a significant portion of grains roll and/or slide along the bed (typical for fluvial NST). Nonetheless, these authors physically derived that, for near-threshold conditions, μ_b can always be approximated as the ratio between only the kinetic contributions, $\mu_b \approx -\langle v'_z v'_x \rangle(0) / \langle v'^2_z \rangle(0)$, and validated this derivation with all their near-threshold simulation data. This insight supports modeling NST across all regimes as a contact-free grain motion (i.e., saltation) above a flat wall, where grain-bed contacts are encoded in the wall boundary conditions.

Appendix C2: Justification for Using Rebound Boundary Conditions

Using rebound boundary conditions for a pure saltation motion is natural, since we are modeling only those grains that have approached a nontrivial steady state trajectory, that is, comparably energetic grains that have survived multiple rebounds with the bed surface without being captured. However, in regard to grains that roll and/or slide, additional justification is needed. A heuristic justification, based on the notion that a rolling regime is equivalent to a regime in which grains perform very small hops (Section C1), is that any grain hopping along the surface will in time experience the entire phase space of possible impact conditions regardless of the typical size of its hops. That is, the impact conditions averaged over sufficiently many impacts are the same for grains performing large hops (saltation) or very small hops (rolling). The only difference is that grains performing large (very small) hops experience the same (varying) statistical impact conditions every single impact. However, we argue that this difference does not matter because we are only interested in modeling the grain dynamics averaged over all fluctuations.

To provide further justification for using rebound boundary conditions, we approximate $\overline{u_x(z)} \approx u_x(\bar{z})$ (Pähz & Durán, 2018a) and rewrite Equation 7 as

$$u_x(\bar{z}) - \bar{v}_x = (\mu_b - S)v_s. \quad (C1)$$

Pähz and Durán (2018a) used equation C1 in the limit of threshold conditions to predict the transport threshold Θ_t . To do this, these authors derived $\bar{z}_t = z_{ct} + v_{z_t}^2 / \bar{g}$, where z_{ct} is the transport layer thickness due to particle contacts (unimportant for our discussion), and combined Equation C1 with Equation 3 and three semiempirical closures:

$$\mu_{bt} = \text{const}, \quad (C2a)$$

$$(\bar{v}_{z_t}^2)^{1/2} \propto \bar{v}_{x_t}, \quad (C2b)$$

$$\frac{\bar{v}_{x_t}}{u_{*t}} = \frac{2}{\kappa} \sqrt{1 - \exp \left[-\frac{1}{4} \kappa^2 \gamma^2 \left(\frac{\bar{u}_{x_t}}{u_{*t}} \right)^2 \right]}, \quad (C2c)$$

where γ is a constant. Using DEM-based simulations of NST for $S = 0$, they found that these closures are roughly valid for near-threshold conditions with seemingly arbitrary s and Ga (i.e., including NST regimes with a significant or predominant rolling motion of grains), with the exception of Equation C2b for *viscous saltation* (i.e., aeolian saltation within the viscous sublayer, a precise definition is provided in Section 3.2), and obtained the values of the parameters that appear in these closures from fitting to their simulation data.

Here, rather than from data fitting, we recover these closures, including the deviation for viscous saltation, from a periodic saltation model with rebound boundary conditions, supporting the use of such a model even for NST regimes with a significant or predominant rolling motion of grains. In fact, it is shown in Section 2.3.2 that equations C2a and C2b correspond to the rebound boundary conditions provided that vertical drag is negligible (i.e., $|a_z^d| \ll \tilde{g}$), which is significantly violated only for viscous saltation (causing slight deviations from Equation C2a and substantial deviations from Equation C2b, see Section 3.2). Furthermore, Pähitz and Durán (2018a) showed that the two extreme regimes in Equation C2c, $\overline{v_{x_t}} \approx \gamma \overline{u_{x_t}}$ for sufficiently small $\overline{u_{x_t}} / u_{*t}$ and $\overline{v_{x_t}} \approx 2\kappa^{-1} u_{*t}$ for sufficiently large $\overline{u_{x_t}} / u_{*t}$, follow from a minimization of Θ . That is, the minimization of Θ in the present model (Equation 16) is the analog of Equation C2c.

Appendix D: Fluid-Particle Interactions

Appendix D1: Drag and Lift

Li et al. (2019) used highly resolved DNS simulations to measure the time series of the bed-tangential (drag) and bed-normal (lift) forces acting on a stationary spherical grain resting in a bed surface pocket or $d/3$ or d above it. That is, these authors measured the total lift force comprised of turbulent lift and shear lift (Dey et al., 2020). They found that this total lift force is negligible relative to the drag force (compare their Figures 9 and 10), except for a grain resting in the pocket. But even in this exceptional case (in most conditions that we model, the grain is most of the time a significant distance above the bed), the average lift force is still only about 1/3 of the drag force. Further contributions to the lift force that would arise if the grain was in motion, such as Magnus lift and centrifugal lift (Dey et al., 2020), are generally substantially smaller than that of the total shear and turbulent lift (Zeng et al., 2009). Owing to the fact that the average lift force rapidly decreases with the distance from the bed (Chepil, 1961) and even becomes negative (Li et al., 2019; Moraga et al., 1999) in a poorly understood fashion, it would be a very difficult task to analytically account for it in a reliable manner even if we chose to not neglect it. Furthermore, even the analytical description of the drag force is associated with substantial uncertainties. For example, standard empirical expressions for the drag force substantially underestimate the actual drag force acting on a spherical grain resting in a bed surface pocket (Li et al., 2019), while the form drag coefficient (C_d^∞ in Equation 6) varies by a factor of about four for typical grain shapes in nature (Camenen, 2007; Raffaele et al., 2020). Given this large uncertainty, it makes little sense to consider additional forces of typically much lower magnitude than the drag force using further and even more uncertain empirical expressions (Section 2.1.2).

Appendix D2: Basset Force

The Basset force depends on a grain's motion history and may become important when the shear Reynolds number Re_d is sufficiently small, like for sand grains in water (Bombardelli et al., 2008). However, its magnitude relative to other fluid-particle interaction forces has been a matter of controversial debate (Lukerchenko, 2010), since the Basset force contribution during a grain-bed interaction tends to compensate the one during the grain's contact-free motion (Lukerchenko, 2012), especially for the steady grain motion considered in the model. For this reason, and since the Basset force is very difficult to be analytically incorporated, it is here neglected.

Appendix D3: Added Mass Force

Pähitz and Durán (2018a) have compared simulations with and without an added mass force term. The results were almost identical, even in the limit $s \rightarrow 1$ where one would have expected otherwise. The reason is the fact that the added mass force is proportional to the sum of the total noncontact force (as it would be in the absence of the added mass effect) and the total contact force, which tend to compensate each other on average when s is close to unity. Hence, if we included the added mass force in the transport threshold model, which does not account for contact forces for reasons explained in Section 2.3, it would only affect the fluid forces and thus create a spurious effect that is not real.

Appendix E: Analytical Solution of Equations 10a–10c and its Approximation

From solving the differential Equations 10a–10c, with the initial condition $\hat{v}(0) = \hat{v}_{\uparrow}$, for $\hat{v}(\hat{t})$, the following expressions are obtained (which can be verified through insertion):

$$\hat{z}(\hat{t}) = (1 + \hat{v}_{\uparrow z})(1 - e^{-\hat{t}}) - \hat{t}, \quad (\text{E1a})$$

$$\hat{v}_z(\hat{t}) = (1 + \hat{v}_{\uparrow z})e^{-\hat{t}} - 1, \quad (\text{E1b})$$

$$\hat{x}(\hat{t}) = \hat{v}_{\uparrow x}(1 - e^{-\hat{t}}) + S(\hat{t} + e^{-\hat{t}} - 1) + \int_0^{\hat{t}} \int_0^{\hat{t}'} \hat{u}_x[\hat{z}(\hat{t}')e^{\hat{t}' - \hat{t}}] e^{\hat{t}' - \hat{t}} d\hat{t}' d\hat{t}, \quad (\text{E1c})$$

$$\begin{aligned} \hat{x}(\hat{t}) \approx & \hat{v}_{\uparrow x}(1 - e^{-\hat{t}}) + S(\hat{t} + e^{-\hat{t}} - 1) \\ & + \hat{u}_x \left\{ \frac{2[3 + \hat{t} + (2 + \hat{t})\hat{v}_{\uparrow z}] - e^{\hat{t}}[6 + \hat{t}^2 + 4\hat{v}_{\uparrow z} - 2\hat{t}(2 + \hat{v}_{\uparrow z})]}{2 + 2e^{\hat{t}}(\hat{t} - 1)} \right\} (\hat{t} + e^{-\hat{t}} - 1), \end{aligned} \quad (\text{E1d})$$

$$\hat{v}_x(\hat{t}) = \hat{v}_{\uparrow x}e^{-\hat{t}} + S(1 - e^{-\hat{t}}) + \int_0^{\hat{t}} \hat{u}_x[\hat{z}(\hat{t}')e^{\hat{t}' - \hat{t}}] e^{\hat{t}' - \hat{t}} d\hat{t}', \quad (\text{E1e})$$

$$\hat{v}_x(\hat{t}) \approx \hat{v}_{\uparrow x}e^{-\hat{t}} + S(1 - e^{-\hat{t}}) + \hat{u}_x \left[2 + \hat{v}_{\uparrow z} - \frac{\hat{t}(1 + \hat{v}_{\uparrow z} + e^{\hat{t}})}{e^{\hat{t}} - 1} \right] (1 - e^{-\hat{t}}). \quad (\text{E1f})$$

To eliminate the integrals, we have used the following approximation in equations E1d and (E1f):

$$\int_0^{\hat{t}} \hat{u}_x[f_1(\hat{t}')] f_2(\hat{t}') d\hat{t}' \approx \hat{u}_x \left[\frac{\int_0^{\hat{t}} f_1(\hat{t}') f_2(\hat{t}') d\hat{t}'}{\int_0^{\hat{t}} f_2(\hat{t}') d\hat{t}'} \right] \int_0^{\hat{t}} f_2(\hat{t}') d\hat{t}', \quad (\text{E2})$$

where the functions $f_1(\hat{t})$ and $f_2(\hat{t})$ stand representative for the functions within the inner integral in equation E1c, within the outer integral in Equation E1c, or within the integral in Equation E1e. This approximation is exact for a linear velocity profile $\hat{u}_x(\hat{z})$ (i.e., within the viscous sublayer) and very accurate within the log-layer, since $f_2(\hat{t})$ changes much more rapidly with $\hat{t}$ than $\hat{u}_x[f_1(\hat{t})]$ in the log-layer (regardless of the integral that is considered).

From Equation E1a and E1b, we obtain the nondimensionalized hop height $\hat{H} \equiv \hat{z}(\hat{v}_z = 0)$ as

$$\hat{H} = \hat{v}_{\uparrow z} - \ln(1 + \hat{v}_{\uparrow z}). \quad (\text{E3})$$

Furthermore, we obtain the nondimensionalized hop time $\hat{T}$ through setting $\hat{z}(\hat{T} > 0) = 0$ in Equation E1a, which yields an implicit expression for $\hat{T}$:

$$\hat{T} = (1 + \hat{v}_{\uparrow z})(1 - e^{-\hat{T}}). \quad (\text{E4})$$

When comparing Equation E4 with Equation E1b evaluated at $\hat{t} = \hat{T}$, we obtain

$$\hat{v}_{\downarrow z} = \hat{v}_{\uparrow z} - \hat{T}. \quad (\text{E5})$$

Moreover, Equation E4 can be rearranged to

$$[\hat{T} - (1 + \hat{v}_{\uparrow z})] e^{\hat{T} - (1 + \hat{v}_{\uparrow z})} = -(1 + \hat{v}_{\uparrow z}) e^{-(1 + \hat{v}_{\uparrow z})}. \quad (\text{E6})$$

Hence, using the definition of the principal branch of the Lambert- W function (i.e., $Y = W(X)$ solves $X = Ye^Y$ for $Y \geq -1$) and $\hat{T} - (1 + \hat{v}_{\uparrow z}) = -\hat{v}_{\downarrow z} - 1 \geq -1$ (Equation E5), we can solve Equation E6 for $\hat{T}$:

$$\hat{T} = 1 + \hat{v}_{\uparrow z} + W \left[-(1 + \hat{v}_{\uparrow z}) e^{-(1 + \hat{v}_{\uparrow z})} \right]. \quad (\text{E7})$$

Put together, Equation E5 and E7 are equivalent to Equation 11.

Evaluating equations E1e and E1f at $\hat{t} = \hat{T}$ yields

$$\hat{v}_{\downarrow x} - \hat{v}_{\uparrow x} = \left\{ -\hat{v}_{\uparrow x} + S + \int_0^{\hat{T}} \hat{u}_x[\hat{z}(\hat{t})] \frac{e^{\hat{t}-\hat{T}}}{1 - e^{-\hat{T}}} d\hat{t} \right\} (1 - e^{-\hat{T}}), \quad (\text{E8a})$$

$$\hat{v}_{\downarrow x} - \hat{v}_{\uparrow x} \approx \left[-\hat{v}_{\uparrow x} + S + \hat{u}_x(\hat{z}_*) \right] (1 - e^{-\hat{T}}), \text{ with } \hat{z}_* \equiv -\hat{v}_{\downarrow z} (1 + \hat{v}_{\uparrow z}) - \hat{v}_{\uparrow z}, \quad (\text{E8b})$$

where we used Equation E4 and E5 to obtain

$$2 + \hat{v}_{\uparrow z} - \frac{\hat{T}(1 + \hat{v}_{\uparrow z} + e^{\hat{T}})}{e^{\hat{T}} - 1} = -\hat{v}_{\downarrow z} (1 + \hat{v}_{\uparrow z}) - \hat{v}_{\uparrow z}. \quad (\text{E9})$$

Furthermore, using Equation 9, E4, and E5 in Equation E8a and E8b yields after some rearrangement:

$$\int_0^{\hat{T}} \hat{u}_x[\hat{z}(\hat{t})] \frac{e^{\hat{t}-\hat{T}}}{1 - e^{-\hat{T}}} d\hat{t} = \mu_b (1 + \hat{v}_{\uparrow z}) + \hat{v}_{\uparrow x} - S, \quad (\text{E10a})$$

$$\hat{u}_x(\hat{z}_*) \approx \mu_b (1 + \hat{v}_{\uparrow z}) + \hat{v}_{\uparrow x} - S. \quad (\text{E10b})$$

It can be easily verified that Equation E10b is equivalent to Equation 12 after inserting Equation 10a.

The average grain velocity-based quantities $\overline{\hat{v}_x}$ and $\hat{v}_z^2$ are obtained from the definition of the time average ($\overline{\cdot} \equiv \frac{1}{\hat{T}} \int_0^{\hat{T}} \cdot d\hat{t}$):

$$\overline{\hat{v}_x} \approx \hat{u}_x(\hat{z}) - \mu_b + S, \quad (\text{E11a})$$

$$\hat{z} = \overline{\hat{v}_z^2} = \frac{1}{2} (\hat{v}_{\uparrow z} + \hat{v}_{\downarrow z}). \quad (\text{E11b})$$

For Equation E11a, we used Equation 7, $\overline{\hat{v}_x} \equiv \overline{\hat{v}_x} / v_s$, and the approximation $\hat{u}_x \approx \hat{u}_x(\hat{z})$ (consistent with Equation E2), while Equation E11b is derived from Equation E1a, E1b, and E5. Note that an alternative approximation for $\overline{\hat{v}_x}$, which yields almost the same values, is given by $\hat{v}_x = \hat{x}(\hat{T}) / \hat{T}$ when using the approximation in Equation E1d to calculate $\hat{x}(\hat{T})$.

Appendix F: Transport Threshold Model in the Limit of Negligible Vertical Drag

The limit of negligible vertical drag (i.e., $|a_z^d| \ll \tilde{g}$) is equivalent to the limit in which grain velocities are much smaller than the terminal settling velocity v_s . Hence, we obtain this limit from Taylor-expanding the transport threshold model in leading order of $\hat{T} = T\tilde{g} / v_s$. For Equations 11 and E5, this implies

$$\hat{v}_{\downarrow z} \approx -\hat{v}_{\uparrow z}, \quad (\text{F1})$$

$$\hat{T} \approx 2\hat{v}_{\uparrow z}, \quad (\text{F2})$$

which means that a grain's kinetic energy is conserved with respect to its vertical motion (i.e., $e_z \equiv -v_{\uparrow z} / v_{\downarrow z} \approx 1$), exactly as one would expect if gravity and buoyancy dominate. Using $e_z \approx 1$, it follows from Equations 13a and 13c that all components of the dimensionless rebound and impact velocities are proportional to $\hat{T}$ and each other in this limit: $\hat{v}_{\uparrow z} \propto \hat{v}_{\downarrow z} \propto \hat{v}_{\uparrow x} \propto \hat{v}_{\downarrow x} \propto \hat{T}$. Hence, the Taylor expansions of Equation E10a and E10b in leading order of $\hat{T}$ read

$$\frac{1}{\hat{T}} \int_0^{\hat{T}} \hat{u}_x[\hat{z}(\hat{t})] d\hat{t} \approx \overline{\hat{u}_x} = \mu_b^o (1 + \hat{v}_{\uparrow z}) + \hat{v}_{\uparrow x} - S, \text{ with } \hat{z}(\hat{t}) \approx \hat{v}_{\uparrow z} \hat{t} - \frac{1}{2} \hat{t}^2, \quad (\text{F3a})$$

$$\hat{u}_x(\hat{z}) \approx \mu_b^o(1 + \hat{v}_{\uparrow z}) + \hat{v}_{\uparrow x} - S, \text{ with } \bar{\hat{z}} = \bar{\hat{v}}_z^2 = \frac{1}{3}\hat{v}_{\uparrow z}^2, \quad (\text{F3b})$$

where μ_b^o can be calculated as (using Equations 13a and 13c)

$$\mu_b^o = \lim_{e_z \rightarrow 1} \mu_b = \frac{1}{2(A+C)^2} \left\{ \sqrt{(1+B+C)^4 - (A+C)^4} - \sqrt{[A(1+B+C)^2 - B(A+C)^2]^2 - (A+C)^4} \right\} = 0.646. \quad (\text{F4})$$

Note that, using Equation E11a, Equation F3a Implies

$$\bar{\hat{v}}_x \approx \mu_b^o \hat{v}_{\uparrow z} + \hat{v}_{\uparrow x} = \frac{1}{2}(\hat{v}_{\uparrow x} + \hat{v}_{\downarrow x}). \quad (\text{F5})$$

Interestingly, in the limit of negligible vertical drag, the dependency of the equations that describe the grain trajectory on the slope number S can be directly linked to the case $S = 0$. To show this, we define $f_S \equiv 1 - S / \mu_b^o$, $\mathbf{v}_{\uparrow(\downarrow)}^* \equiv f_S^{-1/2} \mathbf{v}_{\uparrow(\downarrow)} / \sqrt{s g d}$, $v_{s*}^o \equiv \lim_{e_z \rightarrow 1} v_{s*}$, and the new dimensionless numbers $Ga_S \equiv f_S^{1/2} Ga$, $s_S \equiv f_S s$, and $\Theta_S \equiv f_S^{-1} \Theta$. Then, using the definition of the hat ($\hat{\cdot}$) and Equations 7 and 10a, we can rewrite Equation F3b as

$$\Theta_t^{\text{roll}} \equiv \Theta^{\text{roll}}|_{\Theta=\Theta_t} = 0 \quad (\text{F6})$$

Equation F6 does not contain explicit dependencies on S , implying that the only remaining threshold model equation that contains an explicit dependency on S is the one for Θ^{roll} (Equation 15). Furthermore, for $S = 0$ (i.e., $f_S = 1$), the functional forms of Equation F6 remains unchanged if the modified quantities with subscript S are replaced by the unmodified ones without subscript S . Hence, for saltation conditions ($\Theta_t^{\text{roll}} \equiv \Theta^{\text{roll}}|_{\Theta=\Theta_t} = 0$) in the limit of negligible vertical drag, like for turbulent saltation, Θ_t and $\bar{v}_{x*}$ scale as

$$\Theta_t(Ga, s, S) \approx f_S \Theta_t(\sqrt{f_S} Ga, f_S s, 0), \quad (\text{F7})$$

$$\bar{v}_{x*}(Ga, s, S) \approx \sqrt{f_S} \bar{v}_{x*}(\sqrt{f_S} Ga, f_S s, 0). \quad (\text{F8})$$

It turns out that these scaling laws also work reasonably well for turbulent bedload even though $\Theta_t^{\text{roll}} > 0$ (Figure 7b). Note that, for the derivation of Equations 20a and 20b, we further roughly approximated the right-hand side of these expressions using $\Theta_t(f_S^{1/2} Ga, f_S s, 0) \approx \Theta_t(Ga, s, 0)$ and $\bar{v}_{x*}(f_S^{1/2} Ga, f_S s, 0) \approx \bar{v}_{x*}(Ga, s, 0)$, respectively, since the indirect effects of f_S on Θ_t and $\bar{v}_{x*}$ via rescaling Ga and s are relatively weak.

Appendix G: Explicit Solution of

Equation 23 can be rewritten in the form

$$\Theta_t \approx a[\ln(b\Theta_t) - c]^{-2}, \quad (\text{G1})$$

where $a \equiv [\kappa(\mu_b^o - S)v_{s*}^o]^2$, $b \equiv 30\kappa^{-2}\beta s$, and $c = 2$ for $z_{ot} = d/30$ and $a \equiv [2\kappa(\mu_b^o - S)v_{s*}^o/3]^2$, $b \equiv (9\kappa^{-2}\beta s Ga)^{2/3}$, and $c = 4/3$ for $z_{ot} = d/(9Re_{dt}) = d/(9\Theta_t^{1/2} Ga)$. Note that, using Equations 13a and 13c, the constant β appearing in b can be calculated as

$$\beta = \lim_{e_z \rightarrow 1} \frac{4v_{\uparrow z}^2}{3(\mu_b v_{\uparrow z} + v_{\uparrow x})^2} = \frac{16}{3}(A+C)^4 \left\{ \sqrt{(1+B+C)^4 - (A+C)^4} + \sqrt{[A(1+B+C)^2 - B(A+C)^2]^2 - (A+C)^4} \right\}^{-2} = 0.136. \quad (\text{G2})$$

Now, further rearrangement of Equation G1 successively yields

$$\frac{\ln(b\Theta_t) - c}{2} \exp\left(\frac{\ln(b\Theta_t) - c}{2}\right) = \frac{\sqrt{ab}}{2e^{c/2}}, \quad (\text{G3})$$

$$\frac{\ln(b\Theta_t) - c}{2} \approx W\left(\frac{\sqrt{ab}}{2e^{c/2}}\right), \quad (\text{G4})$$

$$\Theta_t \approx \frac{1}{b} \exp\left[2W\left(\frac{\sqrt{ab}}{2e^{c/2}}\right) + c\right] = \frac{a}{4} W\left(\frac{\sqrt{ab}}{2e^{c/2}}\right)^{-2}, \quad (\text{G5})$$

where we used the definition of the principal branch of the Lambert- W function and the identity $\exp[2W(X)] = [X/W(X)]^2$. The explicit solution for Θ_t in Equation G5 corresponds to Θ_t^{rough} in Equation 24 for $z_{ot} = d/30$ and Θ_t^{smooth} in Equation 24 for $z_{ot} = d/(9Re_{dt})$.

Notation

$\kappa = 0.4$ Von Kármán constant

$C_d^\infty = 0.4$, $m^d = 2$ Constants associated with fluid drag law

$c_M = 1.7$ Model parameter associated with grain fluctuation energy dissipation

z_Δ Distance between rebound elevation and virtual zero-level of flow velocity profile

$Z_\Delta = 0.7 \equiv z_\Delta/d$ (model parameter)

Δz_s Distance between bed crest level and virtual zero-level of flow velocity profile

$A = 0.87$, $B = 0.72$, $C = 0$ Model parameters associated with rebound laws

Θ_Y Yield stress

$\Theta_Y^0 = 0.13$ Yield stress for nonsloped bed ($S = 0$)

ψ Pocket angle of a pocket of the bed surface

$\psi_s = 25^\circ$ Pocket angle of most stable bed surface pocket

ψ_* Pocket angle after kinetic energy of grain located in most stable bed surface pocket has been fully converted into potential energy during initial rolling motion

h_s, h_* Distances depicted in Figure 3

τ_g Particle shear stress

p_g Vertical particle pressure

$\mu_b \equiv \tau_g(0)/p_g(0)$ (bed surface value of friction coefficient)

$\mu_b^0 \equiv \lim_{e_z \rightarrow 1} \mu_b \approx 0.646$

$\beta \equiv \lim_{e_z \rightarrow 1} (4v_{\uparrow z}^2 / 3) / (\mu_b v_{\uparrow z} + v_{\uparrow x})^2 \approx 0.136$

$\mathbf{u}$ Mean flow velocity

m Mass of single grain

$\mathbf{v}$ Grain velocity

$\mathbf{v}'$ Grain fluctuation velocity

$\mathbf{v}_1$ Lift-off or rebound velocity

$\mathbf{v}_1$ Impact velocity

$\mathbf{v}_1^0$ Initial lift-off or rebound velocity

$\mathbf{v}_\uparrow^e$ Average velocity of entrained grain

$E_\uparrow = \frac{1}{2} m \mathbf{v}_\uparrow^2$ (lift-off or rebound grain kinetic energy)

$e \equiv |\mathbf{v}_\downarrow|/|\mathbf{v}_\uparrow|$ (rebound restitution coefficient)

$e_z \equiv -v_{\uparrow z}/v_{\downarrow z}$ (vertical rebound restitution coefficient)

$\theta_\uparrow$ Lift-off or rebound angle

$\theta_\downarrow$ Impact angle

$\mathbf{a}$ Grain acceleration

$\mathbf{a}^d$ Grain acceleration due to fluid drag

$\mathbf{a}^g$ Grain acceleration due to gravity

$\mathbf{a}^b$ Grain acceleration due to buoyancy

$\mathbf{f}_x$ Streamwise fluid-particle interaction force per unit volume

$z_{\max}$ Top of the transport layer, which no grain exceeds

h_m Flow depth, including the sediment-fluid mixture above the bed surface $z = 0$

H Hop height in periodic saltation model

T Hop time in periodic saltation model

ρ Particle concentration

$\rho^\uparrow$ Particle concentration of ascending grains

$\rho^\downarrow$ Particle concentration of descending grains

$\phi^\uparrow$ Vertical upward-flux of grains

$G^\uparrow$ Quantity G during upward-motion of periodic saltation trajectory

$G^\downarrow$ Quantity G during downward-motion of periodic saltation trajectory

$\langle G \rangle$ Local grain mass-weighted ensemble average of quantity G

$\bar{G}$ ρ -weighted height average of quantity G (= time average over grain hop in periodic saltation model)

$\hat{G}$ Quantity G nondimensionalized using units of v_s and $\tilde{g}$

G_t Quantity G in the limit of threshold conditions

v_s^- Terminal grain settling velocity in quiescent flow

v_s Terminal grain settling velocity in equilibrium NST

v_{s*} Nondimensionalized terminal settling velocity of grains in equilibrium NST

$v_{s*}^0 = \lim_{e_c \rightarrow 1} v_{s*}$

d Grain diameter

ρ_p Particle density

ρ_f Fluid density

ν_f Kinematic fluid viscosity

τ Fluid shear stress applied onto bed surface

τ^{fluc} Fluctuating fluid shear stress applied onto bed surface

σ_τ Standard deviation of fluctuating bed fluid shear stress

u_* Fluid shear velocity

α Bed slope angle (slopes aligned with flow direction, for downward slopes, $\alpha > 0$)

α_r Angle of repose

$f_s \equiv 1 - S / \mu_b^0$ (slope correction factor)

$f_\alpha \equiv 1 - \tan \alpha / \mu_b^0$

g Gravity constant

$\tilde{g}$ Vertical buoyancy-reduced value of g

$f_u, f_{\tilde{u}}$ Functions associated with mean flow velocity profile

$Ro \equiv v_s^- / (\kappa u_*)$ (Rouse number, fully NST requires $Ro \gtrsim Ro_c = 2.8$)

$Re_d \equiv u_* d / \nu_f$ (particle Reynolds number)

$Re_z \equiv u_* (z/d + Z_\Delta) / \nu_f$ (wall units)

$s \equiv \rho_p / \rho_f$ (density ratio)

$Ga \equiv d \sqrt{s \tilde{g} d} / \nu_f$ (Galileo number)

$S \equiv -(a_x^g + a_x^b) / (a_z^g + a_z^b)$ (slope number)

$\Theta \equiv \tau / (\rho_p \tilde{g} d)$ (Shields number)

Θ_t Shields number at transport threshold

$\Theta^{\text{fluc}} \equiv \tau^{\text{fluc}} / (\rho_p \tilde{g} d)$

Θ^c Bed surface pocket angle-dependent critical Shields number required for rolling

Θ^{roll} Grain kinetic energy-dependent critical Shields number required for rolling

Θ^{cont} Shields number at continuous transport threshold

Q Sediment transport rate

$Q_* \equiv Q / (\rho_p d \sqrt{s \tilde{g} d})$

M Sediment transport load

$M_* \equiv M / (\rho_p d)$

$\bar{v}_x$ Average streamwise sediment velocity

$\bar{v}_{x*} \equiv \bar{v}_x / \sqrt{s \tilde{g} d}$

W Lambert-W function

Data Availability Statement

All data shown in the figures of this article can be found in the following references: Comola, Gaume, et al. (2019), Pähtz and Durán (2018a), Pähtz and Durán (2020), Meyer-Peter and Müller (1948, pp. 39–64), Karahan (1975), Fernandez Luque and van Beek (1976), Yalin and Karahan (1979), Smart and Jaeggi (1983), Loiseleux et al. (2005), Ouriemi et al. (2007), Capart and Fraccarollo (2011), Bagnold (1937), Chepil (1945), Creyssels et al. (2009), Ho et al. (2011), Ho (2012), Carneiro et al. (2015), Martin and Kok (2017); Martin and Kok (2018), Zhu et al. (2019), Ralaizarisoa et al. (2020), Sugiura et al. (1998), Clifton et al. (2006), and Andreotti et al. (2021).

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